MASTER OF COMPUTER APPLICATIONS

PRACTICAL RECORD WORK
ON
20MCA134 – ADVANCED DBMS LAB

Submitted by

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DEPARTMENT OF COMPUTER APPLICATIONS

COLLEGE OF ENGINEERING VATAKARA

(CAPE – GOVT. OF KERALA)

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DEPARTMENT OF COMPUTER APPLICATIONS

COLLEGE OF ENGINEERING VATAKARA

CAPE(– GOVT. OF KERALA)

CERTIFICATE

Certified that this is the bonafide record work on the practical course **20MCA134 ADVANCED DBMS LAB** done and prepare **Mr. RAHBAR ZAHID S (Reg No.VDA24MCA-2039)**, 2nd Semester **MCA(2024-26 Batch)** student of Departmentof Computer Applications at College of Engineering Vadakara, in the partial fulfilment of the award of MCA Degree of APJ Abdul Kalam Technological University (KTU).

Date :

Faculty-in-Charge **Head of the Department**

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Internal Examiners: **External Examiners:**

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EXPERIMENT NO:1

AIM:

Create database company ,create tables and view tables in the database,

1) Employee(Emp_No, Emp_Name, Address, Sex, Dept, Salary, DOJ, Branch)
2) Department(Dept_No, Dname,Mgr_Id, Mgr_Strtdate)

RESULT:

1) Create a database:

> create database company;
> use company;

```
mysql> create database company;
Query OK, 1 row affected (0.01 sec)

mysql> use company;
Database changed
mysql>
```

2) Create table Employee:

> create table Employee(Emp_No int primary key, Emp_Name varchar(20), Address varchar(20), Sex char(2), Dept varchar(20) ,Salary int, DOJ date,Branch varchar(20));

```
mysql> create table Employee(Emp_No int primary key,Emp_Name varchar(20),Address varchar(20),Sex char(2),Dept varchar(20),Salary int,DOJ dat
e,Branch varchar(20));
Query OK, 0 rows affected (0.04 sec)
```

3)View table Employee:

> desc Employee;

```
mysql> desc Employee;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| Emp_No   | int         | NO   | PRI | NULL    |       |
| Emp_Name | varchar(20) | YES  |     | NULL    |       |
| Address  | varchar(20) | YES  |     | NULL    |       |
| Sex      | char(2)     | YES  |     | NULL    |       |
| Dept     | varchar(20) | YES  |     | NULL    |       |
| Salary   | int         | YES  |     | NULL    |       |
| DOJ      | date        | YES  |     | NULL    |       |
| Branch   | varchar(20) | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
8 rows in set (0.00 sec)
```

4)Create table Department:

> create table Department(Dept_No int primary key ,Dname varchar(20) ,Mgr_Id int ,Mgr_Strtdate date);

```
mysql> create table Department(Dept_No int primary key,Dname varchar(20),Mgr_Id int,Mgr_Strtdate date);
Query OK, 0 rows affected (0.03 sec)
```

5)View table Department:

> desc Department;

```
mysql> desc Department;
+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| Dept_No      | int         | NO   | PRI | NULL    |       |
| Dname        | varchar(20) | YES  |     | NULL    |       |
| Mgr_Id       | int         | YES  |     | NULL    |       |
| Mgr_Strtdate | date        | YES  |     | NULL    |       |
+--------------+-------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

EXPERIMENT NO:2

AIM:
Perform sufficient insertion of records into the database created in the first experiment.
Alter the created table and perform Insertion, Updation and Deletion operation on it.

RESULT:
1)Insert records :

<u>Employee table</u>
> insert into Employee values(1001,'Aysha','Pal','F','HR',34000,'2017-12-02','Finance');
> insert into Employee values(1002,'Arjun','Ern','M','Coding',40000,'2019-10-12','IT');
> insert into Employee values(1003,'Teena','Knr','F','Sales',30000,'2016-12-22','Marketing');
> insert into Employee values(1004,'Rahul','Koz','M','Clerk',25000,'2019-05-15','Finance');
> insert into Employee values(1005,'Aysha','Knr','F','Desgning',35000,'2022-11-25','IT');

```
mysql> insert into Employee values(1001,'Aysha','Pal','F','HR',34000,'2017-12-02','Finance');
Query OK, 1 row affected (0.00 sec)

mysql> insert into Employee values(1002,'Arjun','Ern','M','Coding',40000,'2019-10-12','IT');
Query OK, 1 row affected (0.01 sec)

mysql> insert into Employee values(1003,'Teena','Knr','F','Sales',30000,'2016-12-22','Marketing');
Query OK, 1 row affected (0.00 sec)

mysql> insert into Employee values(1004,'Rahul','Koz','M','Clerk',25000,'2019-05-15','Finance');
Query OK, 1 row affected (0.01 sec)

mysql> insert into Employee values(1005,'Aysha','Knr','F','Desgning',35000,'2022-11-25','IT');
Query OK, 1 row affected (0.01 sec)

mysql> select * from Employee;
+--------+----------+---------+------+----------+--------+------------+-----------+
| Emp_No | Emp_Name | Address | Sex  | Dept     | Salary | DOJ        | Branch    |
+--------+----------+---------+------+----------+--------+------------+-----------+
|   1001 | Aysha    | Pal     | F    | HR       |  34000 | 2017-12-02 | Finance   |
|   1002 | Arjun    | Ern     | M    | Coding   |  40000 | 2019-10-12 | IT        |
|   1003 | Teena    | Knr     | F    | Sales    |  30000 | 2016-12-22 | Marketing |
|   1004 | Rahul    | Koz     | M    | Clerk    |  25000 | 2019-05-15 | Finance   |
|   1005 | Aysha    | Knr     | F    | Desgning |  35000 | 2022-11-25 | IT        |
+--------+----------+---------+------+----------+--------+------------+-----------+
5 rows in set (0.00 sec)
```

<u>Department table</u>
> insert into Department values(101, 'HR', 20063, '2016-12-11'), (102, 'IT', 34063, '2020-11-21'), (103, 'Finance', 52101, '2009-01-13'), (104, 'Sales' ,15201, '2022-07-10') ,(105, 'Finance' ,56564 ,'2022-10-11');

```
mysql> insert into Department values(101,'HR',20063,'2016-12-11'),(102,'IT',34063,'2020-11-21'),(103,'Finance',52101,'2009-01-13'),(104,'Sales',15201,'2022-07-10'),(1
05,'Finance',56564,'2022-10-11');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from Department;
+---------+---------+--------+--------------+
| Dept_No | Dname   | Mgr_Id | Mgr_Strtdate |
+---------+---------+--------+--------------+
|     101 | HR      |  20063 | 2016-12-11   |
|     102 | IT      |  34063 | 2020-11-21   |
|     103 | Finance |  52101 | 2009-01-13   |
|     104 | Sales   |  15201 | 2022-07-10   |
|     105 | Finance |  56564 | 2022-10-11   |
+---------+---------+--------+--------------+
5 rows in set (0.00 sec)
```

2)Update Department table:
> update Department set Mgr_Id=45621 where Dept_No=104;

```
mysql> update Department set Mgr_Id=45621 where Dept_No=104;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from Department;
+---------+---------+--------+--------------+
| Dept_No | Dname   | Mgr_Id | Mgr_Strtdate |
+---------+---------+--------+--------------+
|     101 | HR      |  20063 | 2016-12-11   |
|     102 | IT      |  34063 | 2020-11-21   |
|     103 | Finance |  52101 | 2009-01-13   |
|     104 | Sales   |  45621 | 2022-07-10   |
|     105 | Finance |  56564 | 2022-10-11   |
+---------+---------+--------+--------------+
5 rows in set (0.01 sec)
```

3)Rename the table Department as Dept:
> alter table Department rename to Dept;

```
mysql> alter table Department rename to Dept;
Query OK, 0 rows affected (0.02 sec)

mysql> show tables;
+--------------------+
| Tables_in_company |
+--------------------+
| Dept               |
| Employee           |
+--------------------+
2 rows in set (0.00 sec)
```

4)Add a new column Phone with not null constraints to the existing table Dept:
> alter table Dept add column Phone varchar(20) not null;

```
mysql> alter table Dept add column Phone varchar(20) not null;
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select * from Dept;
+---------+---------+--------+--------------+-------+
| Dept_No | Dname   | Mgr_Id | Mgr_Strtdate | Phone |
+---------+---------+--------+--------------+-------+
|     101 | HR      |  20063 | 2016-12-11   |       |
|     102 | IT      |  34063 | 2020-11-21   |       |
|     103 | Finance |  52101 | 2009-01-13   |       |
|     104 | Sales   |  45621 | 2022-07-10   |       |
|     105 | Finance |  56564 | 2022-10-11   |       |
+---------+---------+--------+--------------+-------+
5 rows in set (0.00 sec)
```

5)Drop column Phone:
> alter table Dept drop Phone;

```
mysql> alter table Dept drop Phone;
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select * from Dept;
+---------+---------+--------+--------------+
| Dept_No | Dname   | Mgr_Id | Mgr_Strtdate |
+---------+---------+--------+--------------+
|     101 | HR      |  20063 | 2016-12-11   |
|     102 | IT      |  34063 | 2020-11-21   |
|     103 | Finance |  52101 | 2009-01-13   |
|     104 | Sales   |  45621 | 2022-07-10   |
|     105 | Finance |  56564 | 2022-10-11   |
+---------+---------+--------+--------------+
5 rows in set (0.00 sec)
```

6) Rename the column DName to Dept_Name in Dept table:
> alter table Dept change column Dname Dept_name varchar(20);

```
mysql> alter table Dept change column Dname Dept_name varchar(20);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select * from Dept;
+---------+-----------+--------+--------------+
| Dept_No | Dept_name | Mgr_Id | Mgr_Strtdate |
+---------+-----------+--------+--------------+
|     101 | HR        |  20063 | 2016-12-11   |
|     102 | IT        |  34063 | 2020-11-21   |
|     103 | Finance   |  52101 | 2009-01-13   |
|     104 | Sales     |  45621 | 2022-07-10   |
|     105 | Finance   |  56564 | 2022-10-11   |
+---------+-----------+--------+--------------+
5 rows in set (0.00 sec)
```

7)Drop table:
>drop table Dept;

```
mysql> select * from Dept;
+---------+---------+--------+--------------+
| Dept_No | Dname   | Mgr_Id | Mgr_Strtdate |
+---------+---------+--------+--------------+
|     101 | HR      |  20063 | 2016-12-11   |
|     102 | IT      |  34063 | 2020-11-21   |
|     103 | Finance |  52101 | 2009-01-13   |
|     104 | Sales   |  45621 | 2022-07-10   |
|     105 | Finance |  56564 | 2022-10-11   |
+---------+---------+--------+--------------+
5 rows in set (0.00 sec)
```

EXPERIMENT NO:3

AIM:

Create database for schemas:

1) Course(Course _ID, Course _Name, Program_ID, Credit ,Semester, Internal_Mark, External_Mark, Course_Type)
2) Student_Mark(Reg_No, Course_ID, Program_ID, Student_Internal, Student_External) After associating these schemas into the already created database and perform row insertion, deletion and updation.

RESULT:

1)Create table course:

> create table course(Course_ID varchar(15) not null primary key, Course_name varchar(20) not null, Program_ID smallint ,Credit smallint, Semester varchar(10), Internal_mark int, External_mark int, Course_Type varchar(10));

```
mysql> create table course(Cousrse_ID varchar(15) not null primary key,Course_name varchar(20) not null,Program_ID smalli
nt,Credit smallint,Semester varchar(10),Internal_mark int,External_mark int,Course_Type varchar(10));
Query OK, 0 rows affected (0.02 sec)
```

2)View table course:

> desc course;

```
mysql> desc course;
+----------------+-------------+------+-----+---------+-------+
| Field          | Type        | Null | Key | Default | Extra |
+----------------+-------------+------+-----+---------+-------+
| Cousrse_ID     | varchar(15) | NO   | PRI | NULL    |       |
| Course_name    | varchar(20) | NO   |     | NULL    |       |
| Program_ID     | smallint    | YES  |     | NULL    |       |
| Credit         | smallint    | YES  |     | NULL    |       |
| Semester       | varchar(10) | YES  |     | NULL    |       |
| Internal_mark  | int         | YES  |     | NULL    |       |
| External_mark  | int         | YES  |     | NULL    |       |
| Course_Type    | varchar(10) | YES  |     | NULL    |       |
+----------------+-------------+------+-----+---------+-------+
8 rows in set (0.00 sec)
```

3)Create table Student_Mark:

>create table Student_Mark(Reg_No varchar(10) primary key, Course_ID varchar(15), Student_Internal int not null, Student_External int not null, foreign key (Course_ID) references course(Course_ID));

```
mysql>  create table Student_Mark(Reg_No varchar(10) primary key,Course_ID varchar(15),Student_Internal int not null,Stud
ent_External int not null, foreign key (Course_ID) references course(Course_ID));
Query OK, 0 rows affected (0.03 sec)
```

4)View table Student_Mark:

> desc Student;

```
mysql> desc Student_Mark;
+------------------+-------------+------+-----+---------+-------+
| Field            | Type        | Null | Key | Default | Extra |
+------------------+-------------+------+-----+---------+-------+
| Reg_No           | varchar(10) | NO   | PRI | NULL    |       |
| Course_ID        | varchar(15) | YES  | MUL | NULL    |       |
| Student_Internal | int         | NO   |     | NULL    |       |
| Student_External | int         | NO   |     | NULL    |       |
+------------------+-------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

5)Insert records:

<u>course table</u>

> insert into course values('mca1', 'MCA', 200, 4, 's2', 45, 55, 'Integrate') ,('mca2', 'MCA', 201, 6, 's4' ,40 ,60, 'Regular') ,('bca3', 'BCA', 202, 8, 's3' ,60, 40, 'Regular'), ('btech4', 'Btech', 203,6, 's4', 50, 50, 'Lateral');

```
mysql> insert into course values('mca1','MCA',200,4,'s2',45,55,'Integrated'),('mca2','MCA',201,6,'s4',40,60,'Regular'),('
bca3','BCA',202,8,'s3',60,40,'Regular'),('btech4','Btech',203,6,'s4',50,50,'Lateral');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> select * from course;
+-----------+-------------+------------+--------+----------+----------------+---------------+-------------+
| Course_ID | Course_name | Program_ID | Credit | Semester | Internal_mark | External_mark | Course_Type |
+-----------+-------------+------------+--------+----------+----------------+---------------+-------------+
| bca3      | BCA         |        202 |      8 | s3       |             60 |            40 | Regular     |
| btech4    | Btech       |        203 |      6 | s4       |             50 |            50 | Lateral     |
| mca1      | MCA         |        200 |      4 | s2       |             45 |            55 | Integrated  |
| mca2      | MCA         |        201 |      6 | s4       |             40 |            60 | Regular     |
+-----------+-------------+------------+--------+----------+----------------+---------------+-------------+
4 rows in set (0.00 sec)

mysql>
```

<u>Student_Mark table</u>

> insert into Student_Mark values('b123' ,'b011', 35, 40), ('m141', 'm111', 28, 42), ('mi768', 'mi313', 34, 52), ('mt836', 'mt811', 55, 48), ('bt67f', 'b0110', 78, 51);

```
mysql> insert into Student_Mark values('b123','b011',35,40),('m141','m111',28,42),('mi768','mi313',34,52),('mt836','mt811
',55,48),('bt67f','b0110',78,51);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
```

6)Display table:

> select * from Student_Mark;

```
mysql> select * from Student_Mark;
+--------+-----------+------------------+------------------+
| Reg_No | Course_ID | Student_Internal | Student_External |
+--------+-----------+------------------+------------------+
| b123   | b011      |               35 |               40 |
| bt67f  | b0110     |               78 |               51 |
| m141   | m111      |               28 |               42 |
| mi768  | mi313     |               34 |               52 |
| mt836  | mt811     |               55 |               48 |
+--------+-----------+------------------+------------------+
5 rows in set (0.00 sec)
```

7)Update:

>update course set Semester='s4' where Course_ID='bca3';

```
mysql> update course set Semester='s4' where Course_ID='bca3';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from course;
+-----------+-------------+------------+--------+----------+----------------+---------------+-------------+
| Course_ID | Course_name | Program_ID | Credit | Semester | Internal_mark | External_mark | Course_Type |
+-----------+-------------+------------+--------+----------+----------------+---------------+-------------+
| bca3      | BCA         |        202 |      8 | s4       |             60 |            40 | Regular     |
| btech4    | Btech       |        203 |      6 | s4       |             50 |            50 | Lateral     |
| mca1      | MCA         |        200 |      4 | s2       |             45 |            55 | Integrated  |
| mca2      | MCA         |        201 |      6 | s4       |             40 |            60 | Regular     |
+-----------+-------------+------------+--------+----------+----------------+---------------+-------------+
4 rows in set (0.00 sec)
```

>update Student_Mark set Student_Internal=28 where Reg_No="bt67f";

```
mysql> update Student_Mark set Student_Internal=28 where Reg_No="bt67f";
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from Student_Mark;
+--------+-----------+------------------+------------------+
| Reg_No | Course_ID | Student_Internal | Student_External |
+--------+-----------+------------------+------------------+
| b123   | b011      |               35 |               40 |
| bt67f  | b0110     |               28 |               51 |
| m141   | m111      |               28 |               42 |
| mi768  | mi313     |               34 |               52 |
| mt836  | mt811     |               55 |               48 |
+--------+-----------+------------------+------------------+
5 rows in set (0.00 sec)
```

8)Rename Course_name to Course_Name in Course Table:

>alter table course change column Course_name Course_Name varchar(20);

```
mysql> alter table course change column Course_name Course_Name varchar(20);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select * from course;
+-----------+-------------+------------+--------+----------+---------------+---------------+-------------+
| Course_ID | Course_Name | Program_ID | Credit | Semester | Internal_mark | External_mark | Course_Type |
+-----------+-------------+------------+--------+----------+---------------+---------------+-------------+
| bca3      | BCA         |        202 |      8 | s4       |            60 |            40 | Regular     |
| btech4    | Btech       |        203 |      6 | s4       |            50 |            50 | Lateral     |
| mca1      | MCA         |        200 |      4 | s2       |            45 |            55 | Integrated  |
| mca2      | MCA         |        201 |      6 | s4       |            40 |            60 | Regular     |
+-----------+-------------+------------+--------+----------+---------------+---------------+-------------+
4 rows in set (0.00 sec)
```

9)Add column Result in Student_Mark table:
> alter table Student_Mark add Result varchar(10);

```
mysql> alter table Student_Mark add Result varchar(10);
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select * from Student_Mark;
+--------+-----------+------------------+------------------+--------+
| Reg_No | Course_ID | Student_Internal | Student_External | Result |
+--------+-----------+------------------+------------------+--------+
| b123   | b011      |               35 |               40 | NULL   |
| bt67f  | b0110     |               28 |               51 | NULL   |
| m141   | m111      |               28 |               42 | NULL   |
| mi768  | mi313     |               34 |               52 | NULL   |
| mt836  | mt811     |               55 |               48 | NULL   |
+--------+-----------+------------------+------------------+--------+
5 rows in set (0.00 sec)
```

10)Drop column Result in Student_mark table:
> alter table Student_Mark drop Result;

```
mysql> alter table Student_Mark drop Result;
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select * from Student_Mark;
+--------+-----------+------------------+------------------+
| Reg_No | Course_ID | Student_Internal | Student_External |
+--------+-----------+------------------+------------------+
| b123   | b011      |               35 |               40 |
| bt67f  | b0110     |               28 |               51 |
| m141   | m111      |               28 |               42 |
| mi768  | mi313     |               34 |               52 |
| mt836  | mt811     |               55 |               48 |
+--------+-----------+------------------+------------------+
5 rows in set (0.00 sec)
```

11)Rename table Student_Mark as std_mark:
> rename table student_mark to std_mark;

```
mysql> rename table Student_Mark to std_mark;
Query OK, 0 rows affected (0.01 sec)

mysql>
```

12)Delete data from std_mark table:
> truncate table course;

```
mysql> truncate table course;
Query OK, 0 rows affected (0.02 sec)

mysql> select * from course;
Empty set (0.00 sec)
```

>drop table course;

```
mysql> drop table course;
Query OK, 0 rows affected (0.01 sec)

mysql> show tables;
+-------------------+
| Tables_in_college |
+-------------------+
| std_mark          |
| student           |
+-------------------+
2 rows in set (0.00 sec)
```

EXPERIMENT NO:4

AIM:
Perform with DCL and TCL commands on the already created table and identifying the changes effected on the table data and control while execution each of the commands.

RESULT:
1) DCL Commands:
GRANT and **REVOKE** commands to assign and revoke privilege.

- **GRANT**

> grant select,insert,update on Employee to Aysha;

```
mysql> create user aysha identified by '@123';
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> grant select,insert,update on Employee1 to aysha;
Query OK, 0 rows affected (0.00 sec)
```

- **REVOKE**

> revoke insert on Employee from Aysha;

```
mysql> revoke insert on Employee1 from aysha;
Query OK, 0 rows affected (0.00 sec)
```

2) TCL Commands:
COMMIT and **ROLLBACK** and **SAVE** commnads to control to change made to the table data.

- **COMMIT**

> insert into Employee1 values(1,'IT'),(2,'Sales)',(3,'Accounts'),(4,'Finance');
> commit;

```
mysql> insert into Employee1 values(1,'IT'),(2,'Sales'),(3,'Accounts'),(4,'Finance');
Query OK, 4 rows affected (0.00 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> commit;
Query OK, 0 rows affected (0.00 sec)
```

- **SAVEPOINT:**
  used to temporarily save a transaction so that we can roll back to that point (saved point) whenever required.

> update Employee1 set Branch = "commerce" where Emp_Id=2;

> savepoint a;

> insert into Employee1 values(7,'Sales');

> savepoint b;

> insert into Employee1 values(8,'Designer');

> savepoint c;

```
mysql> update Employee1 set Branch='Technical' where Emp_Id=2;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> savepoint a;
Query OK, 0 rows affected (0.00 sec)

mysql> insert into Employee1 values(7,'Sales');
Query OK, 1 row affected (0.00 sec)

mysql> savepoint b;
Query OK, 0 rows affected (0.00 sec)

mysql> insert into Employee1 values(8,'Designer');
Query OK, 1 row affected (0.00 sec)

mysql> savepoint c;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from Employee1;
+--------+-----------+
| Emp_Id | branch    |
+--------+-----------+
|      1 | IT        |
|      2 | Technical |
|      3 | Accounts  |
|      4 | Finance   |
|      7 | Sales     |
|      8 | Designer  |
+--------+-----------+
6 rows in set (0.00 sec)
```

ROLLBACK:

Rollback the change made in Transaction

> rollback to insertion;

```
mysql> ROLLBACK TO Insertion;
Query OK, 0 rows affected (0.09 sec)
```

EXPERIMENT NO:5

AIM:

Data retrieval from the already created database in first experiment and then perform sample query selections.

Familiarization of Distinct, All etc.

RESULT:

1)Display all the fields of the Employee table :
> select * from Employee;

```
mysql> select * from Employee;
+--------+----------+-------+------+----------+--------+------------+-----------+
| Emp_No | Emp_Name | Place | Sex  | Dept     | Salary | DOJ        | Branch    |
+--------+----------+-------+------+----------+--------+------------+-----------+
|   1001 | Aysha    | Pal   | F    | HR       |  34000 | 2017-12-02 | Finance   |
|   1002 | Arjun    | Ern   | M    | Coding   |  40000 | 2019-10-12 | IT        |
|   1003 | Teena    | Knr   | F    | Sales    |  30000 | 2016-12-22 | Marketing |
|   1004 | Rahul    | Koz   | M    | Clerk    |  15000 | 2019-05-15 | Finance   |
|   1005 | Aysha    | Knr   | F    | Desgning |  35000 | 2022-11-25 | IT        |
+--------+----------+-------+------+----------+--------+------------+-----------+
5 rows in set (0.00 sec)
```

2) Retrieve employee number and their salary: > select Emp_No, Salary from Employee;

```
mysql> select Emp_No,Salary from Employee;
+--------+--------+
| Emp_No | Salary |
+--------+--------+
|   1001 |  34000 |
|   1002 |  40000 |
|   1003 |  30000 |
|   1004 |  15000 |
|   1005 |  35000 |
+--------+--------+
5 rows in set (0.00 sec)
```

3)Retrieve employee details where salary >20000:
> select * from Employee where Salary >20000;

```
mysql> select * from Employee where Salary>20000;
+--------+----------+-------+------+----------+--------+------------+-----------+
| Emp_No | Emp_Name | Place | Sex  | Dept     | Salary | DOJ        | Branch    |
+--------+----------+-------+------+----------+--------+------------+-----------+
|   1001 | Aysha    | Pal   | F    | HR       |  34000 | 2017-12-02 | Finance   |
|   1002 | Arjun    | Ern   | M    | Coding   |  40000 | 2019-10-12 | IT        |
|   1003 | Teena    | Knr   | F    | Sales    |  30000 | 2016-12-22 | Marketing |
|   1005 | Aysha    | Knr   | F    | Desgning |  35000 | 2022-11-25 | IT        |
+--------+----------+-------+------+----------+--------+------------+-----------+
4 rows in set (0.00 sec)
```

4)Distinct :
> select distinct * from Employee;

```
mysql> select distinct * from Employee;
+--------+----------+-------+------+----------+--------+------------+-----------+
| Emp_No | Emp_Name | Place | Sex  | Dept     | Salary | DOJ        | Branch    |
+--------+----------+-------+------+----------+--------+------------+-----------+
|   1001 | Aysha    | Pal   | F    | HR       |  34000 | 2017-12-02 | Finance   |
|   1002 | Arjun    | Ern   | M    | Coding   |  40000 | 2019-10-12 | IT        |
|   1003 | Teena    | Knr   | F    | Sales    |  30000 | 2016-12-22 | Marketing |
|   1004 | Rahul    | Koz   | M    | Clerk    |  15000 | 2019-05-15 | Finance   |
|   1005 | Aysha    | Knr   | F    | Desgning |  35000 | 2022-11-25 | IT        |
+--------+----------+-------+------+----------+--------+------------+-----------+
5 rows in set (0.00 sec)
```

5)All :
> select all * from Employee;

```
mysql> select all * from Employee;
+--------+----------+-------+------+----------+--------+------------+-----------+
| Emp_No | Emp_Name | Place | Sex  | Dept     | Salary | DOJ        | Branch    |
+--------+----------+-------+------+----------+--------+------------+-----------+
|   1001 | Aysha    | Pal   | F    | HR       |  34000 | 2017-12-02 | Finance   |
|   1002 | Arjun    | Ern   | M    | Coding   |  40000 | 2019-10-12 | IT        |
|   1003 | Teena    | Knr   | F    | Sales    |  30000 | 2016-12-22 | Marketing |
|   1004 | Rahul    | Koz   | M    | Clerk    |  15000 | 2019-05-15 | Finance   |
|   1005 | Aysha    | Knr   | F    | Desgning |  35000 | 2022-11-25 | IT        |
+--------+----------+-------+------+----------+--------+------------+-----------+
5 rows in set (0.00 sec)
```

6)Display details of employee whose name is ‘Arjun’ and salary greater than 15000 :
> select * from Employee where Emp_Name='Arjun' and Salary>15000;

```
mysql> select * from Employee where Emp_Name='Arjun' and Salary>15000;
+--------+----------+-------+------+--------+--------+------------+--------+
| Emp_No | Emp_Name | Place | Sex  | Dept   | Salary | DOJ        | Branch |
+--------+----------+-------+------+--------+--------+------------+--------+
|   1002 | Arjun    | Ern   | M    | Coding |  40000 | 2019-10-12 | IT     |
+--------+----------+-------+------+--------+--------+------------+--------+
1 row in set (0.00 sec)
```

7)Display details of Employee from Kannur and Palakkad :
> select * from Employee where Place in('Knr','Pal');

```
mysql> select * from Employee where Place in('Knr','Pal');
+--------+----------+-------+------+----------+--------+------------+-----------+
| Emp_No | Emp_Name | Place | Sex  | Dept     | Salary | DOJ        | Branch    |
+--------+----------+-------+------+----------+--------+------------+-----------+
|   1001 | Aysha    | Pal   | F    | HR       |  34000 | 2017-12-02 | Finance   |
|   1003 | Teena    | Knr   | F    | Sales    |  30000 | 2016-12-22 | Marketing |
|   1005 | Aysha    | Knr   | F    | Desgning |  35000 | 2022-11-25 | IT        |
+--------+----------+-------+------+----------+--------+------------+-----------+
3 rows in set (0.00 sec)
```

8)Display details of Employee who are not from Kannur and Palakkad :
> select * from Employee where Place not in('Knr','Pal');

```
mysql> select * from Employee where Place not in('Knr','Pal');
+--------+----------+-------+------+--------+--------+------------+---------+
| Emp_No | Emp_Name | Place | Sex  | Dept   | Salary | DOJ        | Branch  |
+--------+----------+-------+------+--------+--------+------------+---------+
|   1002 | Arjun    | Ern   | M    | Coding |  40000 | 2019-10-12 | IT      |
|   1004 | Rahul    | Koz   | M    | Clerk  |  15000 | 2019-05-15 | Finance |
+--------+----------+-------+------+--------+--------+------------+---------+
2 rows in set (0.00 sec)
```

9)Display details of Employee whose salary between 15000 and 30000:
> select * from Employee where Salary between 15000 and 30000;

```
mysql> select * from Employee where Salary between 15000 and 30000;
+--------+----------+-------+------+-------+--------+------------+-----------+
| Emp_No | Emp_Name | Place | Sex  | Dept  | Salary | DOJ        | Branch    |
+--------+----------+-------+------+-------+--------+------------+-----------+
|   1003 | Teena    | Knr   | F    | Sales |  30000 | 2016-12-22 | Marketing |
|   1004 | Rahul    | Koz   | M    | Clerk |  15000 | 2019-05-15 | Finance   |
+--------+----------+-------+------+-------+--------+------------+-----------+
2 rows in set (0.00 sec)
```

10)Display name of employee in descending order :
> select * from Employee order by Emp_Name desc;

```
mysql> select * from Employee order by Emp_Name desc;
+--------+----------+-------+------+----------+--------+------------+-----------+
| Emp_No | Emp_Name | Place | Sex  | Dept     | Salary | DOJ        | Branch    |
+--------+----------+-------+------+----------+--------+------------+-----------+
|   1003 | Teena    | Knr   | F    | Sales    |  30000 | 2016-12-22 | Marketing |
|   1004 | Rahul    | Koz   | M    | Clerk    |  15000 | 2019-05-15 | Finance   |
|   1001 | Aysha    | Pal   | F    | HR       |  34000 | 2017-12-02 | Finance   |
|   1005 | Aysha    | Knr   | F    | Desgning |  35000 | 2022-11-25 | IT        |
|   1002 | Arjun    | Ern   | M    | Coding   |  40000 | 2019-10-12 | IT        |
+--------+----------+-------+------+----------+--------+------------+-----------+
5 rows in set (0.00 sec)
```

EXPERIMENT NO:6

AIM:

Data retrieval from the already created database in third Experiment and do necessary Addition of data then perform sample nested query selections using Comparison and Logical Connectives.

RESULT:

1)display table:

> select * from course;

```
mysql> select * from course;
+-----------+-------------+------------+--------+----------+---------------+---------------+-------------+
| Course_ID | Course_name | Program_ID | Credit | Semester | Internal_mark | External_mark | Course_Type |
+-----------+-------------+------------+--------+----------+---------------+---------------+-------------+
| bca3      | BCA         |        202 |      8 | s3       |            60 |            40 | Regular     |
| btech4    | Btech       |        203 |      6 | s4       |            50 |            50 | Lateral     |
| mca1      | MCA         |        200 |      4 | s2       |            45 |            55 | Integrated  |
| mca2      | MCA         |        201 |      6 | s4       |            40 |            60 | Regular     |
+-----------+-------------+------------+--------+----------+---------------+---------------+-------------+
4 rows in set (0.00 sec)
```

2)Retrieve all student who have score higher than average internal mark in a specific course:

> select * from course where Internal_mark >(select avg(Internal_mark) from course where Course_ID ='bca3');

```
mysql> select * from course where Internal_mark>(select avg(Internal_mark)from course where Course_ID ='bca3');
Empty set (0.00 sec)
```

3)Retrieve all courses with credit greater than 3 and whose semester is either s2 or s4:

> select * from course where Credit >4 and semester in (‘s2’, ‘s4’);

```
mysql> select * from course where Credit>4 and Semester in('s2','s4');
+-----------+-------------+------------+--------+----------+---------------+---------------+-------------+
| Course_ID | Course_name | Program_ID | Credit | Semester | Internal_mark | External_mark | Course_Type |
+-----------+-------------+------------+--------+----------+---------------+---------------+-------------+
| btech4    | Btech       |        203 |      6 | s4       |            50 |            50 | Lateral     |
| mca2      | MCA         |        201 |      6 | s4       |            40 |            60 | Regular     |
+-----------+-------------+------------+--------+----------+---------------+---------------+-------------+
2 rows in set (0.00 sec)
```

4)Retrieve all student who are enrolled in a program with more than 100 students:

> select * from course where Course_ID in(select Course_ID from course where Program_Id=201);

```
mysql> select * from course where Course_ID in(select Course_ID from course where Program_ID=201);
+-----------+-------------+------------+--------+----------+---------------+---------------+-------------+
| Course_ID | Course_name | Program_ID | Credit | Semester | Internal_mark | External_mark | Course_Type |
+-----------+-------------+------------+--------+----------+---------------+---------------+-------------+
| mca2      | MCA         |        201 |      6 | s4       |            40 |            60 | Regular     |
+-----------+-------------+------------+--------+----------+---------------+---------------+-------------+
1 row in set (0.00 sec)
```

5)Retrieve all courses that are not taken by any student:

> select * from course where Credit>(select avg(Credit) from course);

```
mysql> select * from course where Credit>(select avg(Credit)from course);
+-----------+-------------+------------+--------+----------+---------------+---------------+-------------+
| Course_ID | Course_name | Program_ID | Credit | Semester | Internal_mark | External_mark | Course_Type |
+-----------+-------------+------------+--------+----------+---------------+---------------+-------------+
| bca3      | BCA         |        202 |      8 | s3       |            60 |            40 | Regular     |
+-----------+-------------+------------+--------+----------+---------------+---------------+-------------+
1 row in set (0.00 sec)
```

EXPERIMENT NO:7

AIM:

Data retrieval from the already created database/ Create new sample database and do necessary Addition of data then write queries that familiarize all string operations in SQL.

RESULT:

1)Concat

> select concat('hello ','world') as ConcatenatedString;

```
mysql> select concat('hello ','world')as ConcatenatedString;
+---------------------+
| ConcatenatedString |
+---------------------+
| hello world         |
+---------------------+
1 row in set (0.00 sec)
```

2)Substring

> select substring("hello world",7,5) as ExtraString;

```
mysql> select substring("hello world",7,5)as ExtraString;
+-------------+
| ExtraString |
+-------------+
| world       |
+-------------+
1 row in set (0.00 sec)
```

3)Length

> select length("hello world") as LengthOfString;

```
mysql> select length("hello world")as LengthOfString;
+----------------+
| LengthOfString |
+----------------+
|             11 |
+----------------+
1 row in set (0.00 sec)
```

4)Upper

> select upper("hello world");

```
mysql> select upper("hello world");
+-----------------------+
| upper("hello world") |
+-----------------------+
| HELLO WORLD           |
+-----------------------+
1 row in set (0.00 sec)
```

5)Lower

> select lower("HELLO WORLD");

```
mysql> select lower("HELLO WORLD");
+-----------------------+
| lower("HELLO WORLD") |
+-----------------------+
| hello world           |
+-----------------------+
1 row in set (0.00 sec)
```

6) Right Substring

> select right("hello world", 5) as ExtraString;

```
mysql> select right("hello world", 5)as ExtraString;
+-------------+
| ExtraString |
+-------------+
| world       |
+-------------+
1 row in set (0.00 sec)
```

7)Trim

> select trim(" hello world")as TrimmedString;

```
mysql> select trim("    hello world")as TrimmedString;
+---------------+
| TrimmedString |
+---------------+
| hello world   |
+---------------+
1 row in set (0.00 sec)
```

8)Replace

>select replace(“hello world”, ”world” ,”universe”);

```
mysql> select replace("hello world","world","universe");
+-----------------------------------------+
| replace("hello world","world","universe") |
+-----------------------------------------+
| hello universe                          |
+-----------------------------------------+
1 row in set (0.00 sec)
```

EXPERIMENT NO:8

AIM:
Data retrieval from the already created database/ Create new sample database and do necessary Addition of data then write queries that familiarize all aggregate functions, group by and having clauses in SQL.

RESULT:
> select * from course;

```
mysql> select * from course;
+-----------+-------------+-------------+--------+----------+---------------+---------------+-------------+
| Course_ID | Course_name | Program_ID | Credit | Semester | Internal_mark | External_mark | Course_Type |
+-----------+-------------+-------------+--------+----------+---------------+---------------+-------------+
| bca3      | BCA         |         202 |      8 | s3       |            60 |            40 | Regular     |
| btech4    | Btech       |         203 |      6 | s4       |            50 |            50 | Lateral     |
| mca1      | MCA         |         200 |      4 | s2       |            45 |            55 | Integrated  |
| mca2      | MCA         |         201 |      6 | s4       |            40 |            60 | Regular     |
+-----------+-------------+-------------+--------+----------+---------------+---------------+-------------+
4 rows in set (0.00 sec)
```

1)**Count:**
Get number of all courses from course table:
>select count(*) as TotalCourses from course;

```
mysql> select count(*) as TotalCourses from course;
+--------------+
| TotalCourses |
+--------------+
|            4 |
+--------------+
1 row in set (0.00 sec)
```

2)**Sum:**
calculate total credit of all course :
>select sum(Credit) as TotalCredit from course;

```
mysql> select sum(Credit)as TotalCredit from course;
+-------------+
| TotalCredit |
+-------------+
|          24 |
+-------------+
1 row in set (0.00 sec)
```

3)**Average:**
calculate average internal marks of all students:
> select avg(Student_internal) as AverageInternalMark from std_mark;

```
mysql> select avg(Student_internal)as AverageInternalMark from std_mark;
+---------------------+
| AverageInternalMark |
+---------------------+
|             36.0000 |
+---------------------+
1 row in set (0.00 sec)
```

4)**Minimum and Maximum:**
Find the min and max of credit of course :
>select min(Credit) as min,max(Credit) as Max from Course;

```
mysql> select min(Credit) as min,max(Credit)as Max from course;
+------+------+
| min  | Max  |
+------+------+
|    4 |    8 |
+------+------+
1 row in set (0.00 sec)
```

5)**Group By:**

Calculate average internal mark and maximum external mark:
>select Course_ID, avg(Student_internal) as avg, max(Student_internal)as max from std_mark group by Course_ID;

```
mysql> select Course_ID,avg(Student_internal)as avg,max(Student_internal)as max from std_mark group by Course_ID;
+-----------+---------+------+
| Course_ID | avg     | max  |
+-----------+---------+------+
| b011      | 35.0000 |   35 |
| b0110     | 28.0000 |   28 |
| m111      | 28.0000 |   28 |
| mi313     | 34.0000 |   34 |
| mt811     | 55.0000 |   55 |
+-----------+---------+------+
5 rows in set (0.00 sec)
```

6)**Having:**
>select Course_ID, avg(Student_Internal)as avg ,max(Student_External) as max from std_mark group by Course_ID having Course_ID= 'b0110';

```
mysql> select Course_ID,avg(Student_Internal)as avg,max(Student_External)as max from std_mark group by Course_ID having
Course_ID='b0110';
+-----------+---------+------+
| Course_ID | avg     | max  |
+-----------+---------+------+
| b0110     | 28.0000 |   51 |
+-----------+---------+------+
1 row in set (0.00 sec)
```

EXPERIMENT NO :9

AIM:
Data retrieval from the already created database/ Create new sample database and do necessary Addition of data then write queries that familiarize all set operations in SQL.

RESULT:

1)Create database:
> create database onlinestore;

```
mysql> create database onlinestore;
Query OK, 1 row affected (0.01 sec)

mysql> use onlinestore;
Database changed
mysql>
```

2)Create table :
>create table products(id int primary key, name varchar(50), category varchar(50));

```
mysql> create table products(id int primary key,name varchar(50),category varchar(50));
Query OK, 0 rows affected (0.02 sec)

mysql> desc products;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| id       | int         | NO   | PRI | NULL    |       |
| name     | varchar(50) | YES  |     | NULL    |       |
| category | varchar(50) | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

>create table accesories(id int primary key, pname varchar(50), pcategory varchar(50));

```
mysql> create table accesories(id int primary key,pname varchar(50),pcategory varchar(50));
Query OK, 0 rows affected (0.02 sec)

mysql> desc accesories;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| id        | int         | NO   | PRI | NULL    |       |
| pname     | varchar(50) | YES  |     | NULL    |       |
| pcategory | varchar(50) | YES  |     | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

3)Insert Data into table:
> insert into products values(1, ‘phone’, ‘electronics’),
(2,’shirt’,’clothing’),(3,’laptop’,’elevtronics’), (4, ‘shoes’, ‘footwear’), (5, ‘watch’, ‘accessories’);

```
ysql> insert into products values(1,'phone','electronics'),(2,'shirt','clothing'),(3,'laptop','electronics'),(4,'shoes'
'footwear'),(5,'watch','accesories');
uery OK, 5 rows affected (0.01 sec)
ecords: 5  Duplicates: 0  Warnings: 0

ysql> select * from products;
+----+--------+-------------+
| id | name   | category    |
+----+--------+-------------+
  1 | phone  | electronics |
  2 | shirt  | clothing    |
  3 | laptop | electronics |
  4 | shoes  | footwear    |
  5 | watch  | accesories  |
+----+--------+-------------+
 rows in set (0.00 sec)
```

> insert into accesories values(1, 'necklace', 'gold'), (2, 'bangle', 'gold'), (3, 'earing', 'rosegold'), (4, 'ring', 'silver'), (1, 'necklace', 'gold'), (2, 'bangle', 'gold'), (2, 'shirt', 'clothing');

```
mysql> insert into accesories values(1,'necklace','gold'),(2,'bangle','gold'),(3,'earing','rosegold'),(4,'ring','silver'
),(1,'necklace','gold'),(2,'bangle','gold'),(2,'shirt','clothing');
Query OK, 7 rows affected (0.01 sec)
Records: 7  Duplicates: 0  Warnings: 0

mysql> select * from accesories;
+------+----------+-----------+
| id   | pname    | pcategory |
+------+----------+-----------+
|    1 | necklace | gold      |
|    2 | bangle   | gold      |
|    3 | earing   | rosegold  |
|    4 | ring     | silver    |
|    1 | necklace | gold      |
|    2 | bangle   | gold      |
|    2 | shirt    | clothing  |
+------+----------+-----------+
7 rows in set (0.00 sec)
```

4)**Union Operation:**

Retrieve product from different categories:

>select name, category from products where category='electronics' union select name, category from products where category= 'clothing';

```
mysql> select name,category from products where category='electronics' union select name,category from products where ca
tegory='clothing';
+--------+-------------+
| name   | category    |
+--------+-------------+
| phone  | electronics |
| laptop | electronics |
| shirt  | clothing    |
+--------+-------------+
3 rows in set (0.00 sec)
```

> select * from products union select * from accesories;

```
mysql> select * from products union select * from accesories;
+------+----------+-------------+
| id   | name     | category    |
+------+----------+-------------+
|    1 | phone    | electronics |
|    2 | shirt    | clothing    |
|    3 | laptop   | electronics |
|    4 | shoes    | footwear    |
|    5 | watch    | accesories  |
|    1 | necklace | gold        |
|    2 | bangle   | gold        |
|    3 | earing   | rosegold    |
|    4 | ring     | silver      |
+------+----------+-------------+
9 rows in set (0.00 sec)
```

> alter table accesories add price int;
> update accesories set price=22000 where id=1;
> update accesories set price=20000 where id=2 and pcategory= 'gold';
> update accesories set price=16000 where id=3;
> update accesories set price=15000 where id=4;
> update accesories set price=2000 where id=2 and pcategory= 'clothing';

```
mysql> alter table accesories add price int;
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> update accesories set price=22000 where id=1;
Query OK, 2 rows affected (0.01 sec)
Rows matched: 2  Changed: 2  Warnings: 0
```

```
mysql> update accesories set price=16000 where id=3;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update accesories set price=20000 where id=2 and pcategory='gold';
Query OK, 2 rows affected (0.01 sec)
Rows matched: 2  Changed: 2  Warnings: 0

mysql> update accesories set price=15000 where id=4;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> update accesories set price=2000 where id=2 and pcategory='clothing';
Query OK, 1 row affected (0.00 sec)
```

> select * from accessories;

```
mysql> select * from accesories;
+------+----------+-----------+-------+
| id   | pname    | pcategory | price |
+------+----------+-----------+-------+
|    1 | necklace | gold      | 22000 |
|    2 | bangle   | gold      | 20000 |
|    3 | earing   | rosegold  | 16000 |
|    4 | ring     | silver    | 15000 |
|    1 | necklace | gold      | 22000 |
|    2 | bangle   | gold      | 20000 |
|    2 | shirt    | clothing  |  2000 |
+------+----------+-----------+-------+
7 rows in set (0.00 sec)
```

> select id,name,category,null as price from products union select * from accesories;

```
mysql> select id,name,category,null as price from products union select * from accesories;
+------+----------+-------------+-------+
| id   | name     | category    | price |
+------+----------+-------------+-------+
|    1 | phone    | electronics |  NULL |
|    2 | shirt    | clothing    |  NULL |
|    3 | laptop   | electronics |  NULL |
|    4 | shoes    | footwear    |  NULL |
|    5 | watch    | accesories  |  NULL |
|    1 | necklace | gold        | 22000 |
|    2 | bangle   | gold        | 20000 |
|    3 | earing   | rosegold    | 16000 |
|    4 | ring     | silver      | 15000 |
|    2 | shirt    | clothing    |  2000 |
+------+----------+-------------+-------+
10 rows in set (0.00 sec)
```

5) Union all operation :

> select * from products union all select * from accesories;

```
mysql> select * from products union all select * from accesories;
+------+----------+-------------+
| id   | name     | category    |
+------+----------+-------------+
|    1 | phone    | electronics |
|    2 | shirt    | clothing    |
|    3 | laptop   | electronics |
|    4 | shoes    | footwear    |
|    5 | watch    | accesories  |
|    1 | necklace | gold        |
|    2 | bangle   | gold        |
|    3 | earing   | rosegold    |
|    4 | ring     | silver      |
|    1 | necklace | gold        |
|    2 | bangle   | gold        |
|    2 | shirt    | clothing    |
+------+----------+-------------+
12 rows in set (0.00 sec)
```

6) Intersection Operation :

> select * from products intersect select * from accessories;

```
mysql> select * from products intersect select * from accesories;
+----+-------+----------+
| id | name  | category |
+----+-------+----------+
|  2 | shirt | clothing |
+----+-------+----------+
1 row in set (0.00 sec)
```

7) Minus or Except operation :

> select * from products except select * from accesories;

```
mysql> select * from products except select * from accesories;
+----+--------+-------------+
| id | name   | category    |
+----+--------+-------------+
|  1 | phone  | electronics |
|  3 | laptop | electronics |
|  4 | shoes  | footwear    |
|  5 | watch  | accesories  |
+----+--------+-------------+
4 rows in set (0.00 sec)
```

EXPERIMENT NO : 10

AIM :
Define a view on the already created database and create sample view and write sample queries on it.

RESULT :

1)Create view CourseDetails as select course. Course_ID, course.Course_Name , course.Credit , Student_Mark.Student_Internal, Student_Mark.Student_External from course inner join Student_Mark on course.Course_ID = Student_Mark.Course_ID;

```
mysql> CREATE VIEW CourseDetails AS
    -> SELECT Course.Course_ID, Course.Course_Name, Course.Credit, Student_Mark.Student_Internal, Student_Mark.Student_External
    -> FROM Course
    -> INNER JOIN Student_Mark ON Course.Course_ID = Student_Mark.Course_ID;
Query OK, 0 rows affected (0.69 sec)
```

>select * from CourseDetails;

```
mysql> SELECT * FROM CourseDetails;
+-----------+------------------+--------+------------------+------------------+
| Course_ID | Course_Name      | Credit | Student_Internal | Student_External |
+-----------+------------------+--------+------------------+------------------+
|         1 | Computer Science |      3 |            75.00 |            65.00 |
|         2 | English          |      4 |            70.00 |            80.00 |
+-----------+------------------+--------+------------------+------------------+
2 rows in set (0.20 sec)
```

>select * from CourseDetails where course_ID = 1;

```
mysql> SELECT * FROM CourseDetails WHERE Course_ID = 1;
+-----------+------------------+--------+------------------+------------------+
| Course_ID | Course_Name      | Credit | Student_Internal | Student_External |
+-----------+------------------+--------+------------------+------------------+
|         1 | Computer Science |      3 |            75.00 |            65.00 |
+-----------+------------------+--------+------------------+------------------+
1 row in set (0.00 sec)
```

>select Course_ID, AVG(Student_Internal)as AvgInternal Mark, Avg (Student_External)as AvgInternalMark from CourseDetails group by Course_ID;

```
mysql> SELECT Course_ID, AVG(Student_Internal) AS AvgInternalMark, AVG(Student_External) AS AvgExternalMark
    -> FROM CourseDetails
    -> GROUP BY Course_ID;
+-----------+-----------------+-----------------+
| Course_ID | AvgInternalMark | AvgExternalMark |
+-----------+-----------------+-----------------+
|         1 |       75.000000 |       65.000000 |
|         2 |       70.000000 |       80.000000 |
+-----------+-----------------+-----------------+
2 rows in set (0.10 sec)
```

>select Course_Name, Student_Internal from CourseDetails where Credit >3;

```
mysql> SELECT Course_Name, Student_Internal
    -> FROM CourseDetails
    -> WHERE Credit > 3;
+-------------+------------------+
| Course_Name | Student_Internal |
+-------------+------------------+
| English     |            70.00 |
+-------------+------------------+
1 row in set (0.00 sec)
```

EXPERIMENT NO : 11

AIM:

Develop a tiny database system and do necessary adding of data and data retrieval from that.

RESULT:

> create table customer(Customer_ID int primary key, Name varcha r(20), Address varchar(50), Contact_Number varchar(15));

> insert into customer values(1, 'John', '123 main street', '123-456-7890'), (2, 'Jane', '456 Elm street', '987-654-3210'), (3, 'Michal', '789 Oak avenue', '555-123-4567'), (4, 'Emily', '321 Pine road', '222-333-444'), (5, 'Sarah', '567 Maple Lane', '777-888-9999');

> select * from customer;

```
mysql> CREATE TABLE customer1 (    Customer_ID INT PRIMARY KEY,
    ->   Name VARCHAR(20),
    ->   Address VARCHAR(50),
    ->   Contact_Number VARCHAR(15)
    -> );
Query OK, 0 rows affected (1.78 sec)

mysql> INSERT INTO customer1 VALUES (1, 'John Doe', '123 Main Street', '123-456-7890'),(2, 'Jane Smith', '456 Elm Street', '987-654-3210'),(3, 'Michael Johnson', '789 O
ak Avenue', '555-123-4567'),(4, 'Emily Davis', '321 Pine Road', '222-333-4444'),(5, 'Sarah Wilson', '567 Maple Lane', '777-888-9999');
Query OK, 5 rows affected (0.12 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from customer1;
+-------------+----------------+----------------+----------------+
| Customer_ID | Name           | Address        | Contact_Number |
+-------------+----------------+----------------+----------------+
|           1 | John Doe       | 123 Main Street | 123-456-7890   |
|           2 | Jane Smith     | 456 Elm Street  | 987-654-3210   |
|           3 | Michael Johnson | 789 Oak Avenue  | 555-123-4567   |
|           4 | Emily Davis    | 321 Pine Road   | 222-333-4444   |
|           5 | Sarah Wilson   | 567 Maple Lane  | 777-888-9999   |
+-------------+----------------+----------------+----------------+
5 rows in set (0.08 sec)
```

> create table Product1(Product_ID in, primary key, Product_Name varchar(20), Quantity_Per_Unit int, Unit_Price decimal(10,2));

> insert intoProduct1 values(1, 'Rice', 1, 10.99), (2, 'Wheat floor', 1, 8.99) , (3, 'Suger', 1, 5.99), (4, 'oil', 1, 12.99), (5, 'Grains', 1, 2.99);

> select * from Product1;

```
mysql> create table Product1(Product_ID int primary key, Product_Name varchar(20),Quantity_Per_Unit int,Unit_Price decimal(10,2));
Query OK, 0 rows affected (0.02 sec)

mysql> insert into Product1 values(1,'Rice',1,10.99),(2,'Wheat floor',1,8.99),(3,'Suger',1,5.99),(4,'oil',1,12.99),(5,'Grains',1,2.99);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from Product1;
+------------+--------------+-------------------+------------+
| Product_ID | Product_Name | Quantity_Per_Unit | Unit_Price |
+------------+--------------+-------------------+------------+
|          1 | Rice         |                 1 |      10.99 |
|          2 | Wheat floor  |                 1 |       8.99 |
|          3 | Suger        |                 1 |       5.99 |
|          4 | oil          |                 1 |      12.99 |
|          5 | Grains       |                 1 |       2.99 |
+------------+--------------+-------------------+------------+
5 rows in set (0.00 sec)
```

> create table Stock(Stock_ID int primary key, Product_ID int, Available_Quantity int, foreign key(Product_ID) references Product(Product_ID));

>insert into Stock values(1,1,100), (2,2,50),(3,3,200),(4,4,80), (5,5,150);

> select * from Stock;

```
mysql> CREATE TABLE Stock (
    ->   Stock_ID INT PRIMARY KEY,
    ->   Product_ID INT,
    ->   Available_Quantity INT,
    ->   FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)
    -> );
Query OK, 0 rows affected (2.39 sec)

mysql> INSERT INTO Stock VALUES (1, 1, 100), (2, 2, 50), (3, 3, 200),(4, 4, 80), (5, 5, 150);
Query OK, 5 rows affected (0.40 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from stock ;
+----------+------------+--------------------+
| Stock_ID | Product_ID | Available_Quantity |
+----------+------------+--------------------+
|        1 |          1 |                100 |
|        2 |          2 |                 50 |
|        3 |          3 |                200 |
|        4 |          4 |                 80 |
|        5 |          5 |                150 |
+----------+------------+--------------------+
5 rows in set (0.00 sec)
```

>create table Transaction(Transaction_ID int primary key, Customer_ID int, Product_ID int, Quantity int, Total_Price decimal(10, 2), Transaction_Date date, foreign key(Customer_ID) references Customer(Customer_ID), foreign key(Product_ID) references Product1(Product_ID));

>insert into Transaction values(1,1,1,2,21.98,’2023-05-24’), (2,2,2,3,26.97,’2023-05-25’), (3,3,3,1,5.99,'2023-05-26'), (4,4,4,5,64.95,'2023-05-27'), (5,5,5,2,5.98,'2023-05-28');

> select * from Transaction;

```
mysql> CREATE TABLE Transaction (
    ->   Transaction_ID INT PRIMARY KEY,
    ->   Customer_ID INT,
    ->   Product_ID INT,
    ->   Quantity INT,
    ->   Total_Price DECIMAL(10, 2),
    ->   Transaction_Date DATE,
    ->   FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID),
    ->   FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)
    -> );
Query OK, 0 rows affected (1.52 sec)

mysql> INSERT INTO Transaction VALUES (1, 1, 1, 2, 21.98, '2023-05-24'), (2, 2, 2, 3, 26.97, '2023-05-25'), (3, 3, 3, 1, 5.99, '2023-05-26'), (4, 4, 4, 5, 64.95, '2023-
05-27'), (5, 5, 5, 2, 5.98, '2023-05-28');
Query OK, 5 rows affected (0.22 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from Transaction ;
+----------------+-------------+------------+----------+-------------+------------------+
| Transaction_ID | Customer_ID | Product_ID | Quantity | Total_Price | Transaction_Date |
+----------------+-------------+------------+----------+-------------+------------------+
|              1 |           1 |          1 |        2 |       21.98 | 2023-05-24       |
|              2 |           2 |          2 |        3 |       26.97 | 2023-05-25       |
|              3 |           3 |          3 |        1 |        5.99 | 2023-05-26       |
|              4 |           4 |          4 |        5 |       64.95 | 2023-05-27       |
|              5 |           5 |          5 |        2 |        5.98 | 2023-05-28       |
+----------------+-------------+------------+----------+-------------+------------------+
5 rows in set (0.00 sec)
```

>select Product_Name,Available_Quantity from Stock inner join Product1 on Stock.Product_ID=Product1.Product_ID where Product_Name = 'Rice';

```
mysql> SELECT Product_Name, Available_Quantity FROM Stock
    -> INNER JOIN Product1 ON Stock.Product_ID = Product1.Product_ID
    -> WHERE Product_Name = 'Rice';
+--------------+--------------------+
| Product_Name | Available_Quantity |
+--------------+--------------------+
| Rice         |                100 |
+--------------+--------------------+
1 row in set (0.10 sec)
```

>insert into Transaction values(6,1,1,5,250,’2023-05-24’);

```
ysql> INSERT INTO Transaction  VALUES (6,1, 1, 5, 250, '2023-05-24');
uery OK, 1 row affected (0.18 sec)

ysql> select * from Transaction ;
+----------------+-------------+------------+----------+-------------+------------------+
| Transaction_ID | Customer_ID | Product_ID | Quantity | Total_Price | Transaction_Date |
+----------------+-------------+------------+----------+-------------+------------------+
|              1 |           1 |          1 |        2 |       21.98 | 2023-05-24       |
|              2 |           2 |          2 |        3 |       26.97 | 2023-05-25       |
|              3 |           3 |          3 |        1 |        5.99 | 2023-05-26       |
|              4 |           4 |          4 |        5 |       64.95 | 2023-05-27       |
|              5 |           5 |          5 |        2 |        5.98 | 2023-05-28       |
|              6 |           1 |          1 |        5 |      250.00 | 2023-05-24       |
+----------------+-------------+------------+----------+-------------+------------------+
 rows in set (0.00 sec)
```

>select Transaction_Id, Product_Name, Quantity, Total_Price, Transaction_Date from Transaction inner join Product1 on Transaction.Product_ID = Product1.Product_ID where Customer_ID=1;

```
mysql> SELECT Transaction_ID, Product_Name, Quantity, Total_Price, Transaction_Date FROM Transaction
    -> INNER JOIN Product1 ON Transaction.Product_ID = Product1.Product_ID
    -> WHERE Customer_ID = 1;
+----------------+--------------+----------+-------------+------------------+
| Transaction_ID | Product_Name | Quantity | Total_Price | Transaction_Date |
+----------------+--------------+----------+-------------+------------------+
|              1 | Rice         |        2 |       21.98 | 2023-05-24       |
|              6 | Rice         |        5 |      250.00 | 2023-05-24       |
+----------------+--------------+----------+-------------+------------------+
2 rows in set (0.10 sec)
```

EXPERIMENT NO: 12

AIM:

Creates a table named students to store student data. Create an procedure for inserting data into student table.

RESULT:

```
CREATE TABLE students (
        id INT PRIMARY KEY,
        name VARCHAR(100),
        department VARCHAR(50)
       );

DELIMITER //
       CREATE PROCEDURE add_student (
          IN p_id INT,
           IN p_name VARCHAR(100),
         IN p_dept VARCHAR(50)
       )
       BEGIN
         INSERT INTO students (id, name, department)
         VALUES (p_id, p_name, p_dept);
        END //
```

- CALL add_student(1, 'Anu', 'CSE');//

```
mysql> use mca
Database changed
mysql> CREATE TABLE students (
    ->     id INT PRIMARY KEY,
    ->     name VARCHAR(100),
    ->     department VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE add_student (
    ->     IN p_id INT,
    ->     IN p_name VARCHAR(100),
    ->     IN p_dept VARCHAR(50)
    -> )
    -> BEGIN
    ->     INSERT INTO students (id, name, department)
    ->     VALUES (p_id, p_name, p_dept);
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> CALL add_student(1, 'Anu', 'CSE');
    -> //
Query OK, 1 row affected (0.03 sec)

mysql> select * from students;//
+----+------+------------+
| id | name | department |
+----+------+------------+
|  1 | Anu  | CSE        |
+----+------+------------+
1 row in set (0.00 sec)
```

EXPERIMENT NO: 13

AIM:

Library Book Management database schema contains book_det(bid, btitle,copies) book_issue(bid,sid,btitle). Create trigger book_copies_deducts For, if any student borrows a book from library then the count of that specified book should be decremented

RESULT:

```
CREATE TABLE book_det (
   bid INT PRIMARY KEY,
   btitle VARCHAR(100),
   copies INT
);
CREATE TABLE book_issue (
   bid INT,
   sid INT,
   btitle VARCHAR(100)
);
CREATE TRIGGER book_copies_deducts
AFTER INSERT ON book_issue FOR EACH ROW
BEGIN
   UPDATE book_det
   SET copies = copies - 1
   WHERE bid = NEW.bid; END;
```

```
mysql> CREATE TABLE book_det (
    ->      bid INT PRIMARY KEY,
    ->      btitle VARCHAR(100),
    ->      copies INT
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE book_issue (
    ->      bid INT,
    ->      sid INT,
    ->      btitle VARCHAR(100)
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TRIGGER book_copies_deducts
    -> AFTER INSERT ON book_issue
    -> FOR EACH ROW
    -> UPDATE book_det
    -> SET copies = copies - 1
    -> WHERE bid = NEW.bid;
Query OK, 0 rows affected (0.01 sec)

mysql> INSERT INTO book_det (bid, btitle, copies)
    -> VALUES (101, 'Wings of Fire', 3),
    ->        (102, 'Harry Potter', 5);
Query OK, 2 rows affected (0.02 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> INSERT INTO book_issue (bid, sid, btitle)
    -> VALUES (101, 1, 'Wings of Fire');
Query OK, 1 row affected (0.01 sec)

mysql> select * from book_det;
+-----+---------------+--------+
| bid | btitle        | copies |
+-----+---------------+--------+
| 101 | Wings of Fire |      2 |
| 102 | Harry Potter  |      5 |
+-----+---------------+--------+
2 rows in set (0.00 sec)
```

EXPERIMENT NO: 14

AIM:

Create table Employee and create view and perform update and delete view.

RESULT

CREATE TABLE Employee (
id INT, name VARCHAR(50),
department VARCHAR(50),salary INT
);
INSERT INTO Employee (id, name, department, salary)
VALUES(1, 'Alice', 'HR', 30000),(2, 'Bob', 'IT', 40000),
(3, 'Carol', 'HR', 35000),(4, 'David', 'Finance', 45000);

- CREATE VIEW HR_Employees AS SELECT id, name, salary FROM Employee WHERE department = 'HR';
- SELECT * FROM HR_Employees;

```
mysql> CREATE TABLE Employee (
    ->     id INT,
    ->     name VARCHAR(50),
    ->     department VARCHAR(50),
    ->     salary INT
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> INSERT INTO Employee (id, name, department, salary)
    -> VALUES
    -> (1, 'Alice', 'HR', 30000),
    -> (2, 'Bob', 'IT', 40000),
    -> (3, 'Carol', 'HR', 35000),
    -> (4, 'David', 'Finance', 45000);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> CREATE VIEW HR_Employees AS
    -> SELECT id, name, salary
    -> FROM Employee
    -> WHERE department = 'HR';
Query OK, 0 rows affected (0.03 sec)

mysql> SELECT * FROM HR_Employees;
+------+-------+--------+
| id   | name  | salary |
+------+-------+--------+
|    1 | Alice |  30000 |
|    3 | Carol |  35000 |
+------+-------+--------+
2 rows in set (0.00 sec)
```

- CREATE OR REPLACE VIEW HR_Employees AS SELECT id, name, department, salary FROM Employee WHERE department = 'HR';
- DROP VIEW HR_Employees;

```
mysql> CREATE OR REPLACE VIEW HR_Employees AS SELECT id, name, department, salary
    -> FROM Employee WHERE department = 'HR';
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM HR_Employees;
+------+-------+------------+--------+
| id   | name  | department | salary |
+------+-------+------------+--------+
|    1 | Alice | HR         |  30000 |
|    3 | Carol | HR         |  35000 |
+------+-------+------------+--------+
2 rows in set (0.00 sec)

mysql> DROP VIEW HR_Employees;
Query OK, 0 rows affected (0.02 sec)
```

EXPERIMENT NO: 1

AIM:

1. Create a database (Eg : MyCev)
2. Create a collection (Eg: db_mca)
3. Create a collection (Eg: db_cs)
4. Insert 10 data to the collection
5. List the first 5 data from the collection (limit)
6. List the entire data except first 2 data (skip)
7. Sort the data by choosing any field in the collection
8. Delete data from the collection
9. Drop the collection (db_cs)
10. Drop Database

RESULT:

1) Create a database (Eg : MyCev) >
use MyCev

```
test> use MyCev
switched to db MyCev
MyCev>
```

2) Create a collection (Eg: db_mca) >
db.createCollection("db_mca")

```
MyCev> db.createCollection("db_mca")
{ ok: 1 }
```

3) Create a collection (Eg: db_cs) >
db.createCollection("db_cs")

```
MyCev> db.createCollection("db_cs")
{ ok: 1 }
MyCev>
```

4)Insert 10 data to the collection

> db.db_mca.insertMany([{"reg_no": 1, "name": "Adhish", "age": 21}, {"reg_no": 2, "name": "Abi", "age": 22}, {"reg_no": 3, "name": "Adith", "age": 20}, {"reg_no": 4, "name": "Ali", "age": 21}, {"reg_no": 5, "name": "Joju", "age": 22}, {"reg_no": 6, "name": "Nithin", "age": 21}, {"reg_no": 7, "name": "Sonu", "age": 20}, {"reg_no": 8, "name": "Ashwin", "age": 22}, {"reg_no": 9, "name": "Benher", "age": 23}, {"reg_no": 10, "name": "Jibin", "age": 21}])

```
MyCev> db.db_mca.insertMany([
...     {"reg_no": 1, "name": "Adhish", "age": 21},
...     {"reg_no": 2, "name": "Abi", "age": 22},
...     {"reg_no": 3, "name": "Adith", "age": 20},
...     {"reg_no": 4, "name": "Ali", "age": 21},
...     {"reg_no": 5, "name": "Joju", "age": 22},
...     {"reg_no": 6, "name": "Nithin", "age": 21},
...     {"reg_no": 7, "name": "Sonu", "age": 20},
...     {"reg_no": 8, "name": "Ashwin", "age": 22},
...     {"reg_no": 9, "name": "Benher", "age": 23},
...     {"reg_no": 10, "name": "Jibin", "age": 21}
... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('666ecb805db854c5e0cdce00'),
    '1': ObjectId('666ecb805db854c5e0cdce01'),
    '2': ObjectId('666ecb805db854c5e0cdce02'),
    '3': ObjectId('666ecb805db854c5e0cdce03'),
    '4': ObjectId('666ecb805db854c5e0cdce04'),
    '5': ObjectId('666ecb805db854c5e0cdce05'),
    '6': ObjectId('666ecb805db854c5e0cdce06'),
    '7': ObjectId('666ecb805db854c5e0cdce07'),
    '8': ObjectId('666ecb805db854c5e0cdce08'),
    '9': ObjectId('666ecb805db854c5e0cdce09')
  }
}
MyCev> |
```

4) List the first 5 data from the collection (limit) > db.db_mca.find().limit(5)

```
MyCev> db.db_mca.find().limit(5)
[
  {
    _id: ObjectId('666ecb805db854c5e0cdce00'),
    reg_no: 1,
    name: 'Adhish',
    age: 21
  },
  {
    _id: ObjectId('666ecb805db854c5e0cdce01'),
    reg_no: 2,
    name: 'Abi',
    age: 22
  },
  {
    _id: ObjectId('666ecb805db854c5e0cdce02'),
    reg_no: 3,
    name: 'Adith',
    age: 20
  },
  {
    _id: ObjectId('666ecb805db854c5e0cdce03'),
    reg_no: 4,
    name: 'Ali',
    age: 21
  },
  {
    _id: ObjectId('666ecb805db854c5e0cdce04'),
    reg_no: 5,
    name: 'Joju',
    age: 22
  }
]
```

5) List the entire data except first 2 data (skip)
> db.db_mca.find().skip(2)

```
MyCev> db.db_mca.find().skip(2)
[
  {
    _id: ObjectId('666ecb805db854c5e0cdce02'),
    reg_no: 3,
    name: 'Adith',
    age: 20
  },
  {
    _id: ObjectId('666ecb805db854c5e0cdce03'),
    reg_no: 4,
    name: 'Ali',
    age: 21
  },
  {
    _id: ObjectId('666ecb805db854c5e0cdce04'),
    reg_no: 5,
    name: 'Joju',
    age: 22
  },
  {
    _id: ObjectId('666ecb805db854c5e0cdce05'),
    reg_no: 6,
    name: 'Nithin',
    age: 21
  },
  {
    _id: ObjectId('666ecb805db854c5e0cdce06'),
    reg_no: 7,
    name: 'Sonu',
    age: 20
  },
  {
    _id: ObjectId('666ecb805db854c5e0cdce07'),
    reg_no: 8,
    name: 'Ashwin',
    age: 22
  },
  {
    _id: ObjectId('666ecb805db854c5e0cdce08'),
    reg_no: 9,
    name: 'Benher',
    age: 23
  },
  {
    _id: ObjectId('666ecb805db854c5e0cdce09'),
    reg_no: 10,
    name: 'Jibin',
    age: 21
  }
]
MyCev>
```

6) Sort the data by choosing any field in the collection.

> db.db_mca.find().sort({age:1})

```
MyCev> db.db_mca.find().sort({age:1})
[
  {
    _id: ObjectId('666ecb805db854c5e0cdce02'),
    reg_no: 3,
    name: 'Adith',
    age: 20
  },
  {
    _id: ObjectId('666ecb805db854c5e0cdce06'),
    reg_no: 7,
    name: 'Sonu',
    age: 20
  },
  {
    _id: ObjectId('666ecb805db854c5e0cdce00'),
    reg_no: 1,
    name: 'Adhish',
    age: 21
  },
  {
    _id: ObjectId('666ecb805db854c5e0cdce03'),
    reg_no: 4,
    name: 'Ali',
    age: 21
  },
  {
    _id: ObjectId('666ecb805db854c5e0cdce05'),
    reg_no: 6,
    name: 'Nithin',
    age: 21
  },
  {
    _id: ObjectId('666ecb805db854c5e0cdce09'),
    reg_no: 10,
    name: 'Jibin',
    age: 21
  },
  {
    _id: ObjectId('666ecb805db854c5e0cdce01'),
    reg_no: 2,
    name: 'Abi',
    age: 22
  },
  {
    _id: ObjectId('666ecb805db854c5e0cdce04'),
    reg_no: 5,
    name: 'Joju',
    age: 22
  },
  {
    _id: ObjectId('666ecb805db854c5e0cdce07'),
    reg_no: 8,
    name: 'Ashwin',
    age: 22
  },
  {
    _id: ObjectId('666ecb805db854c5e0cdce08'),
    reg_no: 9,
    name: 'Benher',
    age: 23
  }
]
```

7) Delete data from the collection

> db.db_mca.deleteOne({name:"Adith"})

```
MyCev> db.db_mca.deleteOne({name:"Adith"})
{ acknowledged: true, deletedCount: 1 }
MyCev>
```

8) Drop the collection
(db_cs) > db.db_cs.drop()

```
MyCev> db.db_cs.drop()
true
```

9) Drop Database
> db.dropDatabase()

```
MyCev> db.dropDatabase()
{ ok: 1, dropped: 'MyCev' }
```

EXPERIMENT NO: 2

AIM:

I. Create a database Myclass
II. Create a collection named “db_students”
   ➔ Should contain this fields : { student_name, student_rollno, mark [subject, mark] }

BI. Insert details of 10 students in a class IV. List the entire students in the class
V. Update mark of any students in the collection “db_students” VI. Delete the data of first student in the collection

RESULT:

1) Create a database
Myclass > use MyClass

```
Myclass> use MyClass
switched to db MyClass
```

2) Create a collection named “db_students” >
db.createCollection("db_students")

```
MyClass> db.createCollection("db_students")
{ ok: 1 }
```

3) Insert details of 10 students in a class
> db.db_students.insertMany([{"student_name":"Adhish","student_rollno":1,"mark":[{" subject ":"Physics","mark":36},{"subject":"Chemistry","mark":46},{"subject":"Maths","mark":29}]},{"student_name":"Ali","student_rollno":2,"mark":[{"subject":"Physics","mark":45},{"subject":"Chemistry","mark":36},{"subject":"Maths","mark":24}]},{"student_name":"Abi","student_rollno":3,"mark":[{"subject":"Physics","mark":56},{"subject":"Chemistry","mark":66 },{"subject":"Maths","mark":59}]},{"student_name":"Benher","student_rollno":4,"mark":[{"subject":"Physics","mark":26},{"subject":"Chemistry","mark":89},{"subject":"Maths","mark":59}]},{"student_name":"Adith","student_rollno":5,"mark":[{"subject":"Physics","mark":86},{"subject":"Chemistry","mark":76},{"subject":"Maths","mark":89}]},{"student_name":"Ashwin","student_rollno":6,"mark":[{"subject":"Physics","mark":88},{"subject":"Chemistry","mark":91 },{"subject":"Maths","mark":45}]},{"student_name":"Sonu","student_rollno":7,"mark":[{"subject":"Physics","mark":65},{"subject":"Chemistry","mark":87},{"subject":"Maths","mark":79}]},{"student_name":"Jibin","student_rollno":8,"mark":[{"subject":"Physics","mark":45},{" subject":"Chemistry","mark":46},{"subject":"Maths","mark":79}]},{"student_name":" Sooraj","student_rollno":9,"mark":[{"subject":"Physics","mark":56},{"subject":"Chemistry","mark":48 },{"subject":"Maths","mark":69}]},{"student_name":"Yedhu","student_rollno":10,"mark":[{"subject":"Physics","mark":49},{"subject":"Chemistry","mark":79},{"subject":"Maths","mark": 69}]})

```
MyClass> db.db_students.insertMany([{"student_name":"Adhish","student_rollno":1,"mark":[{"subject":"Physics","m
ark":36},{"subject":"Chemistry","mark":46},{"subject":"Maths","mark":29}]},{"student_name":"Ali","student_rolln
o":2,"mark":[{"subject":"Physics","mark":45},{"subject":"Chemistry","mark":36},{"subject":"Maths","mark":24}]},
{"student_name":"Abi","student_rollno":3,"mark":[{"subject":"Physics","mark":56},{"subject":"Chemistry","mark":
66},{"subject":"Maths","mark":59}]},{"student_name":"Benher","student_rollno":4,"mark":[{"subject":"Physics","m
ark":26},{"subject":"Chemistry","mark":89},{"subject":"Maths","mark":59}]},{"student_name":"Adith","student_rol
lno":5,"mark":[{"subject":"Physics","mark":86},{"subject":"Chemistry","mark":76},{"subject":"Maths","mark":89}]
},{"student_name":"Ashwin","student_rollno":6,"mark":[{"subject":"Physics","mark":88},{"subject":"Chemistry","m
ark":91},{"subject":"Maths","mark":45}]},{"student_name":"Sonu","student_rollno":7,"mark":[{"subject":"Physics"
,"mark":65},{"subject":"Chemistry","mark":87},{"subject":"Maths","mark":79}]},{"student_name":"Jibin","student_
rollno":8,"mark":[{"subject":"Physics","mark":45},{"subject":"Chemistry","mark":46},{"subject":"Maths","mark":7
9}]},{"student_name":"Sooraj","student_rollno":9,"mark":[{"subject":"Physics","mark":56},{"subject":"Chemistry"
,"mark":48},{"subject":"Maths","mark":69}]},{"studensubject":"Physics","mark":49},{"subject":"Chemistry","mark"
:79},{"subject":"Maths","mark":69}]}])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('666ed29f5db854c5e0cdce14'),
    '1': ObjectId('666ed29f5db854c5e0cdce15'),
    '2': ObjectId('666ed29f5db854c5e0cdce16'),
    '3': ObjectId('666ed29f5db854c5e0cdce17'),
    '4': ObjectId('666ed29f5db854c5e0cdce18'),
    '5': ObjectId('666ed29f5db854c5e0cdce19'),
    '6': ObjectId('666ed29f5db854c5e0cdce1a'),
    '7': ObjectId('666ed29f5db854c5e0cdce1b'),
    '8': ObjectId('666ed29f5db854c5e0cdce1c'),
    '9': ObjectId('666ed29f5db854c5e0cdce1d')
  }
}
```

4) List the entire students in the class >db.db_students.find()

```
MyClass> db.db_students.find()
[
  {
    _id: ObjectId('6618e2253a14f2c8ca16c9b7'),
    student_name: 'Adhish',
    student_rollno: 1,
    mark: [
      { subject: 'Physics', mark: 36 },
      { subject: 'Chemistry', mark: 46 },
      { subject: 'Maths', mark: 29 }
    ]
  },
  {
    _id: ObjectId('6618e2253a14f2c8ca16c9b8'),
    student_name: 'Ali',
    student_rollno: 2,
    mark: [
      { subject: 'Physics', mark: 45 },
      { subject: 'Chemistry', mark: 36 },
      { subject: 'Maths', mark: 24 }
    ]
  },
  {
    _id: ObjectId('6618e2253a14f2c8ca16c9b9'),
    student_name: 'Abi',
    student_rollno: 3,
    mark: [
      { subject: 'Physics', mark: 56 },
      { subject: 'Chemistry', mark: 66 },
      { subject: 'Maths', mark: 59 }
    ]
  },
```

```
  {
    _id: ObjectId('6618e2253a14f2c8ca16c9ba'),
    student_name: 'Benher',
    student_rollno: 4,
    mark: [
      { subject: 'Physics', mark: 26 },
      { subject: 'Chemistry', mark: 89 },
      { subject: 'Maths', mark: 59 }
    ]
  },
  {
    _id: ObjectId('6618e2253a14f2c8ca16c9bb'),
    student_name: 'Adith',
    student_rollno: 5,
    mark: [
      { subject: 'Physics', mark: 86 },
      { subject: 'Chemistry', mark: 76 },
      { subject: 'Maths', mark: 89 }
    ]
  },
  {
    _id: ObjectId('6618e2253a14f2c8ca16c9bc'),
    student_name: 'Ashwin',
    student_rollno: 6,
    mark: [
      { subject: 'Physics', mark: 88 },
      { subject: 'Chemistry', mark: 91 },
      { subject: 'Maths', mark: 45 }
    ]
  },
  {
    _id: ObjectId('6618e2253a14f2c8ca16c9bd'),
    student_name: 'Sonu',
    student_rollno: 7,
    mark: [
      { subject: 'Physics', mark: 65 },
```

```
      { subject: 'Chemistry', mark: 87 },
      { subject: 'Maths', mark: 79 }
    ]
  },
  {
    _id: ObjectId('666ed26b5db854c5e0cdce11'),
    student_name: 'Jibin',
    student_rollno: 8,
    mark: [
      { subject: 'Physics', mark: 45 },
      { subject: 'Chemistry', mark: 46 },
      { subject: 'Maths', mark: 79 }
    ]
  },
  {
    _id: ObjectId('666ed26b5db854c5e0cdce12'),
    student_name: 'Sooraj',
    student_rollno: 9,
    mark: [
      { subject: 'Physics', mark: 56 },
      { subject: 'Chemistry', mark: 48 },
      { subject: 'Maths', mark: 69 }
    ]
  },
  {
    _id: ObjectId('666ed26b5db854c5e0cdce13'),
    student_name: 'Yedhu',
    student_rollno: 10,
    mark: [
      { subject: 'Physics', mark: 49 },
      { subject: 'Chemistry', mark: 79 },
      { subject: 'Maths', mark: 69 }
    ]
  }
]
```

5) Update mark of any students in the collection “db_students”

> db.db_students.updateOne({ "student_name": "Yedhu", "mark.subject": "Physics" }, { $set: { "mark.$.mark": 40 } })

```
MyClass> db.db_students.updateOne(
...     { "student_name": "Yedhu", "mark.subject": "Physics" },
...     { $set: { "mark.$.mark": 40 } }
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}
```

```
  {
    _id: ObjectId('666ed26b5db854c5e0cdce13'),
    student_name: 'Yedhu',
    student_rollno: 10,
    mark: [
      { subject: 'Physics', mark: 40 },
      { subject: 'Chemistry', mark: 79 },
      { subject: 'Maths', mark: 69 }
    ]
  }
]
```

6) Delete the data of first student in the collection

> db.db_students.deleteOne({"student_rollno":1})

```
MyClass> db.db_students.deleteOne({ "student_rollno": 1 })
{ acknowledged: true, deletedCount: 1 }
```

EXPERIMENT NO: 3

AIM:

Create the following collections and its documents in the database LIBRARY

Collection: BOOK

book_id	bktitle	bkedition	bkauthor	bkcost	bkcopies
101,	Database Management Systems ,	7,	Navathe,	450.00	10
102,	Cloud Computing,	2,	Sehgal,	750.00	7
103,	Software Engineering,	5,	Roger	210.00	10
104,	Database Concepts,	5,	Henry	520.00	5

Collection :STUDENT

name	regno	region
SMITH,	12345,	TVM
ALLEN,	56789,	CLT
JEN,	67890,	CLT
BLAKE,	78901,	EKM
WARD,	89012,	CLT
JAMES,	90123,	EKM

RESULT

- Creating database LIBRARY
  > use LIBRARY;

```
test> use LIBRARY
switched to db LIBRARY
```

- Creating Collection BOOK and STUDENT and inserting data to the collections
  > db.createCollection("BOOK");
  > db.BOOK.insertMany([{"book_id":101,"bktitle":"Database Management Systems","bkeditio":7,"bkauthor":"Navathe","bkcost":450.00,"bkcopies":10},{"book_id":102,"bktitle":"CloudComputing","bkedition":2,"bkauthor":"Sehgal","bkcost":750.00,"bkcopies":7},{"book_id":103 ,"bktitle":"SoftwareEngineering","bkedition":5,"bkautho":"Roger","bkcost":210.00,"bkcopies":10},{"book_id":104,"bktitle":"DatabaseConcepts","bkedition":5,"bkauthor":"Henry","bkcost ":520.00,"bkcopies":5}])

```
LIBRARY> db.createCollection("BOOK");
{ ok: 1 }
LIBRARY> db.BOOK.insertMany([{"book_id":101,"bktitle":"Database Management Systems","bkedition":7,"bkauthor":"Navathe","bkcost":45
0.00,"bkcopies":10},{"book_id":102,"bktitle":"Cloud Computing","bkedition":2,"bkauthor":"Sehgal","bkcost":750.00,"bkcopies":7},{"b
ook_id":103,"bktitle":"Software Engineering","bkedition":5,"bkauthor":"Roger","bkcost":210.00,"bkcopies":10},{"book_id":104,"bktit
le":"Database Concepts","bkedition":5,"bkauthor":"Henry","bkcost":520.00,"bkcopies":5}]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('666ef1d430b467f3f4cdcdf6'),
    '1': ObjectId('666ef1d430b467f3f4cdcdf7'),
    '2': ObjectId('666ef1d430b467f3f4cdcdf8'),
    '3': ObjectId('666ef1d430b467f3f4cdcdf9')
  }
}
```

> db.createCollection("STUDENT");

> db.STUDENT.insertMany([{"name":"SMITH","regno":12345,"region":"TVM"},{"name":"ALLEN","regno":56789,"region":"CLT"},{"name":"JEN","regno":67890,"region":"CLT"},{"name":"BLAKE","regno":78901,"region":"EKM"},{"name":"WARD","regno":89012,"region":"CLT"},{"name":"JAMES","regno":90123,"region":"EKM"}])

```
LIBRARY> db.createCollection("STUDENT");
{ ok: 1 }
LIBRARY> db.STUDENT.insertMany([{"name":"SMITH","regno":12345,"region":"TVM"},{"name":"ALLEN","regno":56789,"region":"CLT"},{"name
":"JEN","regno":67890,"region":"CLT"},{"name":"BLAKE","regno":78901,"region":"EKM"},{"name":"WARD","regno":89012,"region":"CLT"},{
"name":"JAMES","regno":90123,"region":"EKM"}]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('666ef39c30b467f3f4cdcdfa'),
    '1': ObjectId('666ef39c30b467f3f4cdcdfb'),
    '2': ObjectId('666ef39c30b467f3f4cdcdfc'),
    '3': ObjectId('666ef39c30b467f3f4cdcdfd'),
    '4': ObjectId('666ef39c30b467f3f4cdcdfe'),
    '5': ObjectId('666ef39c30b467f3f4cdcdff')
  }
}
```

EXPERIMENT NO: 4

AIM:

Write down the MongoDB queries for the following questions.

a) Update the region as ‘TVM’ for the student ‘JAMES’.
b) Display all student documents who reside in the city ‘EKM’ or ‘TVM’.
c) Display student details in the ascending order of their names.

RESULT:

a) Update the region as 'TVM' for the student 'JAMES'.

> db.STUDENT.updateOne({ "name": "JAMES" }, { $set: { "region": "TVM" } });

```
LIBRARY> db.STUDENT.updateOne( { "name": "JAMES" }, { $set: { "region": "TVM" } } );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
LIBRARY> db.STUDENT.find({ "name": "JAMES" })
[
  {
    _id: ObjectId('666ee34b5c2eb62f96cdcdff'),
    name: 'JAMES',
    regno: 90123,
    region: 'TVM'
  }
]
```

b) Display all student documents who reside in the city ‘EKM’ or ‘TVM’.

> db.student.find({ "region": { $in: ["EKM", "TVM"] } });

```
LIBRARY> db.STUDENT.find({ "region": { $in: ["EKM", "TVM"] } });
[
  {
    _id: ObjectId('666ee34b5c2eb62f96cdcdfa'),
    name: 'SMITH',
    regno: 12345,
    region: 'TVM'
  },
  {
    _id: ObjectId('666ee34b5c2eb62f96cdcdfd'),
    name: 'BLAKE',
    regno: 78901,
    region: 'EKM'
  },
  {
    _id: ObjectId('666ee34b5c2eb62f96cdcdff'),
    name: 'JAMES',
    regno: 90123,
    region: 'TVM'
  }
]
```

c) Display student details in the ascending order of their names.

> db.student.find().sort({ "name": 1 });

```
LIBRARY> db.STUDENT.find().sort({ "name": 1 });
[
  {
    _id: ObjectId('666ee34b5c2eb62f96cdcdfb'),
    name: 'ALLEN',
    regno: 56789,
    region: 'CLT'
  },
  {
    _id: ObjectId('666ee34b5c2eb62f96cdcdfd'),
    name: 'BLAKE',
    regno: 78901,
    region: 'EKM'
  },
  {
    _id: ObjectId('666ee34b5c2eb62f96cdcdff'),
    name: 'JAMES',
    regno: 90123,
    region: 'TVM'
  },
  {
    _id: ObjectId('666ee34b5c2eb62f96cdcdfc'),
    name: 'JEN',
    regno: 67890,
    region: 'CLT'
  },
  {
    _id: ObjectId('666ee34b5c2eb62f96cdcdfa'),
    name: 'SMITH',
    regno: 12345,
    region: 'TVM'
  },
  {
    _id: ObjectId('666ee34b5c2eb62f96cdcdfe'),
    name: 'WARD',
    regno: 89012,
    region: 'CLT'
  }
]
```

EXPERIMENT NO: 5

AIM:

Create the following collections and documents in the database BOOKREVIEW

Collection: USER

u_id	u_name	location	dob
101,	SMITH,	TVM	2008-10-03
102,	ALLEN,	CLT	2000-11-23,
103,	JONES,	CLT	2001-10-28,
104,	JAMES,	EKM	2001-03-11,

Collection: BOOK

b_id	b_nmae	author
201,	dbms,	navathe,
202,	cloud computing,	Sehgal,
203,	operating systems,	silberschatz,
204,	software engineering,	Roger,
205,	android,	JohnHorton,
206,	python,	Guido Van

RESULT

- Creating database
  BOOKREVIEW > use
  BOOKREVIEW;

```
test> use BOOKREVIEW
switched to db BOOKREVIEW
```

-
- Creating Collection USER and BOOK and inserting data to the collections
  > db.createCollection("USER");
  > db.USER.insertMany([{"u_id":101,"u_name":"SMITH","location":"TVM","dob":"200 8-10- 03 "},{"u_id":102,"u_name":"ALLEN","location":"CLT","dob":"2000-10-23"},{"u_id":103,"u_name ":"JONES","location":"CLT","dob":"2001-10-28"},{"u_id":104,"u_name":"JAMES","location":" EKM","dob":"2001-03-11"}]);

```
BOOKREVIEW> db.createCollection("USER");
{ ok: 1 }
BOOKREVIEW> db.USER.insertMany([{"u_id":101,"u_name":"SMITH","location":"TVM","dob":"2008-10-03"},{"u_id":102,"u_name":"ALLEN","lo
cation":"CLT","dob":"2000-10-23"},{"u_id":103,"u_name":"JONES","location":"CLT","dob":"2001-10-28"},{"u_id":104,"u_name":"JAMES","
location":"EKM","dob":"2001-03-11"}]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('666ef5d830b467f3f4cdce04'),
    '1': ObjectId('666ef5d830b467f3f4cdce05'),
    '2': ObjectId('666ef5d830b467f3f4cdce06'),
    '3': ObjectId('666ef5d830b467f3f4cdce07')
  }
}
```

> db.createCollection("BOOK");

> db.BOOK.insertMany([{"b_id":201,"b_name":"DBMS","author":"navathe"},{"b_id":202, "b_name ":"cloud computing","author":"sehgal"},{"b_id":203,"b_name":"operating system","author":" silberschatz"},{"b_id":204,"b_name":"software engineering","author":"roger"},{"b_id":205,"b_ name":"android","authorid":"john horton"},{"b_id":206,"b_name":"python","author": "Guido van"}]);

```
BOOKREVIEW> db.createCollection("BOOK");
{ ok: 1 }
BOOKREVIEW> db.BOOK.insertMany([{"b_id":201,"b_name":"DBMS","author":"navathe"},{"b_id":202,"b_name":"cloud computing","author":"s
ehgal"},{"b_id":203,"b_name":"operating system","author":"silberschatz"},{"b_id":204,"b_name":"software engineering","author":"rog
er"},{"b_id":205,"b_name":"android","authorid":"john horton"},{"b_id":206,"b_name":"python","author":"Guido van"}]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('666ef65730b467f3f4cdce08'),
    '1': ObjectId('666ef65730b467f3f4cdce09'),
    '2': ObjectId('666ef65730b467f3f4cdce0a'),
    '3': ObjectId('666ef65730b467f3f4cdce0b'),
    '4': ObjectId('666ef65730b467f3f4cdce0c'),
    '5': ObjectId('666ef65730b467f3f4cdce0d')
  }
}
```

EXPERIMENT NO: 6

AIM:

Write down the MongoDB queries for the following questions.

a) Update the book name ‘android’ as ‘ANDROID OS’.

b) Display book details of ‘dbms’ and ‘python’.

c) Display book details in the descending order of their author names.

RESULT

a) Update the book name ‘android’ as ‘ANDROID OS’.

> db.book.updateOne({ "b_name": "android" }, { $set: { "b_name": "ANDROID OS" } });

```
BOOKREVIEW> db.BOOK.updateOne({ "b_name": "android" }, { $set: { "b_name": "ANDROID OS" } });
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
BOOKREVIEW> db.BOOK.find()
[
  {
    _id: ObjectId('666ef072e9177ef7a3cdcdfa'),
    b_id: 201,
    b_name: 'DBMS',
    author: 'navathe'
  },
  {
    _id: ObjectId('666ef072e9177ef7a3cdcdfb'),
    b_id: 202,
    b_name: 'cl oud computing',
    author: 'sehgal'
  },
  {
    _id: ObjectId('666ef072e9177ef7a3cdcdfc'),
    b_id: 203,
    b_name: 'operating system',
    author: 'silberschatz'
  },
  {
    _id: ObjectId('666ef072e9177ef7a3cdcdfd'),
    b_id: 204,
    b_name: 'software engineering',
    author: 'roger'
  },
```

```
  {
    _id: ObjectId('666ef65730b467f3f4cdce0c'),
    b_id: 205,
    b_name: 'android',
    authorid: 'john horton'
  },
  {
    _id: ObjectId('666ef65730b467f3f4cdce0d'),
    b_id: 206,
    b_name: 'python',
    author: 'Guido van'
  }
]
```

b) Display book details of ‘dbms’ and ‘python’.

> db.BOOK.find({$or:[{"b_name":"DBMS"},{"b_name":"python"}]});

```
BOOKREVIEW> db.BOOK.find({$or:[{"b_name":"DBMS"},{"b_name":"python"}]});
[
  {
    _id: ObjectId('666ef072e9177ef7a3cdcdfa'),
    b_id: 201,
    b_name: 'DBMS',
    author: 'navathe'
  },
  {
    _id: ObjectId('666ef65730b467f3f4cdce0d'),
    b_id: 206,
    b_name: 'python',
    author: 'Guido van'
  }
]
```

c) Display book details in the descending order of their author names. > db.book.find().sort({ "author": -1 });

```
BOOKREVIEW> db.book.find().sort({ "author": -1 });
[
  {
    _id: ObjectId('666efb8b30b467f3f4cdce10'),
    b_id: 203,
    b_name: 'operating systems',
    author: 'Silberschtz'
  },
  {
    _id: ObjectId('666efb8b30b467f3f4cdce0f'),
    b_id: 202,
    b_name: 'cloud computing',
    author: 'Sehgal'
  },
  {
    _id: ObjectId('666efb8b30b467f3f4cdce11'),
    b_id: 204,
    b_name: 'software engineering',
    author: 'Roger'
  },
  {
    _id: ObjectId('666efb8b30b467f3f4cdce0e'),
    b_id: 201,
    b_name: 'dbms',
    author: 'Navathe'
  },
  {
    _id: ObjectId('666efb8b30b467f3f4cdce12'),
    b_id: 205,
    b_name: 'android',
    author: 'John Horton'
  },
  {
    _id: ObjectId('666efb8b30b467f3f4cdce13'),
    b_id: 206,
    b_name: 'python',
    author: 'Guido Van'
  }
]
```

EXPERIMENT NO: 7

AIM:
Create the following collections and documents in the database GRADING

Collection: STUDENT

reg_no	Fname	Lname	Address	Sub_name	credits
501,	SMITH	FORD	HSNO:121,	dbms,	8
502,	ALLEN		HSNO:122	dbms	5
503,	JONES	BLAKE	HSNO:123	sw engineering	9
504,	JAMES	JEORGE	HSNO:124	sw engineering	7
505,	JAMES		HSNO:124	dbms	4

Collection: COURSE

c_code	c_name
201,	MCA,
202,	MBA
203,	B-TECH
204,	M-TECH

RESULT :

- Creating database
  GRADING >use
  GRADING;

```
test> use GRADING
switched to db GRADING
```

- Creating Collection STUDENT and COURSE and inserting data to the collections
  > db.createCollection("STUDENT");
  > db.STUDENT.insertMany([{"reg_no":501,"fname":"SMITH","lname":"FO
  RD","Address":"HSNO:121","Sub_name":"dbms","Credit":8},{"reg_no":50
  2,"fname":"ALLEN","Address":"HSNO:122","Sub_name":"dbms","Credit":
  5},{"reg_no":503," fname":"JONES","lname":"BLAKE","Address":"HS
  NO:123","Sub_name":"swengineering","Credit":9},{"reg_no":504,"fname":"
  JAMES","lname":"JEORGE","Address":"HSNO:124","Sub_name":"swengi
  neering","Credit":7},{"reg_no":505,"fname":"JAMES","Address":"HSNO:1
  24","Sub_name":" dbms","Credit":4}]);

```
GRADING> db.createCollection("STUDENT");
{ ok: 1 }
GRADING> db.STUDENT.insertMany([{"reg_no":501,"fname":"SMITH","lname":"FORD","Address":"HSNO:121","Sub_name":"dbms","Credit":8},{"
reg_no":502,"fname":"ALLEN","Address":"HSNO:122","Sub_name":"dbms","Credit":5},{"reg_no":503,"fname":"JONES","lname":"BLAKE","Addr
ess":"HSNO:123","Sub_name":"swengineering","Credit":9},{"reg_no":504,"fname":"JAMES","lname":"JEORGE","Address":"HSNO:124","Sub_na
me":"swengineering","Credit":7},{"reg_no":505,"fname":"JAMES","Address":"HSNO:124","Sub_name":"dbms","Credit":4}]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('666eff363a951f5ef0cdcdf6'),
    '1': ObjectId('666eff363a951f5ef0cdcdf7'),
    '2': ObjectId('666eff363a951f5ef0cdcdf8'),
    '3': ObjectId('666eff363a951f5ef0cdcdf9'),
    '4': ObjectId('666eff363a951f5ef0cdcdfa')
  }
}
```

> db.createCollection("COURSE");

> db.COURSE.insertMany([{"c_code":201,"c_name":"MCA"},{"c_code":202,"c_name":" MBA"},{" c_code": 203, "c_name":"B-TECH"},{"c_code":204,"c_name":"M-TECH"}]);

```
GRADING> db.createCollection("COURSE");
{ ok: 1 }
GRADING> db.COURSE.insertMany([{"c_code":201,"c_name":"MCA"},{"c_code":202,"c_name":"MBA"},{"c_code":203,"c_name":"B-TECH"},{"c_co
de":204,"c_name":"M-TECH"}]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('666effdd3a951f5ef0cdcdfb'),
    '1': ObjectId('666effdd3a951f5ef0cdcdfc'),
    '2': ObjectId('666effdd3a951f5ef0cdcdfd'),
    '3': ObjectId('666effdd3a951f5ef0cdcdfe')
  }
}
```

EXPERIMENT NO: 8

AIM:

Write down the MongoDB queries for the following questions.

a) Display all documents from the collection 'STUDENT' using pretty().
b) Find minimum and maximum credit of each subjects.
c) Display student details except 'JAMES'.

RESULT :

a) Display all documents from the collection 'STUDENT' using pretty().

> db.STUDENT.find().pretty();

```
GRADING> db.STUDENT.find().pretty();
[
  {
    _id: ObjectId('666eff363a951f5ef0cdcdf6'),
    reg_no: 501,
    fname: 'SMITH',
    lname: 'FORD',
    Address: 'HSNO:121',
    Sub_name: 'dbms',
    Credit: 8
  },
  {
    _id: ObjectId('666eff363a951f5ef0cdcdf7'),
    reg_no: 502,
    fname: 'ALLEN',
    Address: 'HSNO:122',
    Sub_name: 'dbms',
    Credit: 5
  },
  {
    _id: ObjectId('666eff363a951f5ef0cdcdf8'),
    reg_no: 503,
    fname: 'JONES',
    lname: 'BLAKE',
    Address: 'HSNO:123',
    Sub_name: 'swengineering',
    Credit: 9
  },
  {
    _id: ObjectId('666eff363a951f5ef0cdcdf9'),
    reg_no: 504,
    fname: 'JAMES',
    lname: 'JEORGE',
    Address: 'HSNO:124',
    Sub_name: 'swengineering',
    Credit: 7
  },
  {
    _id: ObjectId('666eff363a951f5ef0cdcdfa'),
    reg_no: 505,
    fname: 'JAMES',
    Address: 'HSNO:124',
    Sub_name: 'dbms',
    Credit: 4
  }
]
```

b) Find minimum and maximum credit of each subjects.

```
> db.STUDENT.aggregat
   e ([ { "$group": {
      "_id": "$Sub_name",
      "minCredit": { "$min": "$Credit" },
      "maxCredit": { "$max": "$Credit" }
   }}
]);
```

```
GRADING> db.STUDENT.aggregate([
...     {  "$group": {
...         "_id": "$Sub_name",
...         "minCredit": { "$min": "$Credit" },
...         "maxCredit": { "$max": "$Credit" }
...     }}
... ]);
[
  { _id: 'dbms', minCredit: 4, maxCredit: 8 },
  { _id: 'swengineering', minCredit: 7, maxCredit: 9 }
]
```

c) Display student details except ‘JAMES’.

```
> db.STUDENT.find( { fname: { $ne: "JAMES" } } );
```

```
GRADING> db.STUDENT.find({ "fname": { "$ne": "JAMES" } });
[
  {
    _id: ObjectId('666eff363a951f5ef0cdcdf6'),
    reg_no: 501,
    fname: 'SMITH',
    lname: 'FORD',
    Address: 'HSNO:121',
    Sub_name: 'dbms',
    Credit: 8
  },
  {
    _id: ObjectId('666eff363a951f5ef0cdcdf7'),
    reg_no: 502,
    fname: 'ALLEN',
    Address: 'HSNO:122',
    Sub_name: 'dbms',
    Credit: 5
  },
  {
    _id: ObjectId('666eff363a951f5ef0cdcdf8'),
    reg_no: 503,
    fname: 'JONES',
    lname: 'BLAKE',
    Address: 'HSNO:123',
    Sub_name: 'swengineering',
    Credit: 9
  }
]
```

EXPERIMENT NO: 9

AIM:

Create the following collections and documents in the database MARKETING

Collection: EMPLOYEE

emp_id	emp_name	address	location
101,	SMITH	HNO:123	CLT
102,	ALLEN	HNO:456	EKM
103,	JONES	HNO:789	EKM
104,	JAMES	HNO:321	TVM

Collection: PRODUCT

p_id	p_name	stock	price
201,	laptop,	50,	40000.00
202,	mobile phone,	70,	25000.00
203,	pen drive,	25,	1000.00
204,	smart watch,	100,	30000.00

RESULT :

- Creating database MARKETING > use MARKETING

```
test> use MARKETING
switched to db MARKETING
```

- Creating Collection EMPLOYEE and PRODUCT and inserting data to the collections

>db.createCollection("EMPLOYEE");

> db.EMPLOYEE.insertMany([{"emp_id":101,"emp_name":"SMITH","address":"HNO:123","location":"CLT" },{"emp_id":102,"emp_name":"ALLEN","address":"HNO:456","location":"EKM"},{"emp_id":103,"emp_name":"JONES","address":"HNO:789","location":"EKM"},{"emp_id":104,"emp_name":"JAMES","address" :"HNO:321","location":"TVM"}]);

```
MARKETING> db.createCollection("EMPLOYEE");
{ ok: 1 }
MARKETING> db.EMPLOYEE.insertMany([{"emp_id":101,"emp_name":"SMITH","address":"HNO:123","location":"CLT"},{"emp_id":102,"emp_name"
:"ALLEN","address":"HNO:456","location":"EKM"},{"emp_id":103,"emp_name":"JONES","address":"HNO:789","location":"EKM"},{"emp_id":10
4,"emp_name":"JAMES","address":"HNO:321","location":"TVM"}]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('666f0907b839dad0a1cdcdf6'),
    '1': ObjectId('666f0907b839dad0a1cdcdf7'),
    '2': ObjectId('666f0907b839dad0a1cdcdf8'),
    '3': ObjectId('666f0907b839dad0a1cdcdf9')
  }
}
```

> db.createCollection("PRODUCT");

> db.PRODUCT.insertMany([{"p_id":201,"p_name":"laptop","stock":50,"price":4000 0.00 },{"p_id":202,"p_name":"mobilephone","stock":70,"price":25000.00},{"p_id": 203,"p_name":"pendrive","stock":25,"price":1000.00},{"p_id":204,"p-name":"smart watch"," stock ":100,"price":30000.00}]);

```
MARKETING> db.createCollection("PRODUCT");
{ ok: 1 }
MARKETING> db.PRODUCT.insertMany([{"p_id":201,"p_name":"laptop","stock":50,"price":40000.00},{"p_id":202,"p_name":"mobile phone","
stock":70,"price":25000.00},{"p_id":203,"p_name":"pendrive","stock":25,"price":1000.00},{"p_id":204,"p_name":"smart watch","stock"
:100,"price":30000.00}]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('666f09a7b839dad0a1cdcdfa'),
    '1': ObjectId('666f09a7b839dad0a1cdcdfb'),
    '2': ObjectId('666f09a7b839dad0a1cdcdfc'),
    '3': ObjectId('666f09a7b839dad0a1cdcdfd')
  }
}
```

EXPERIMENT NO: 10

AIM:

Write down the MongoDB queries for the following questions.

a. Update the location ‘EKM’ as ‘ERN’

b. Display employee details in ascending order of their names.

c. Display product names whose price greater than 25000.

RESULT :

a) Update the location ‘EKM’ as ‘ERN’

> db.EMPLOYEE.updateMany({ "location": "EKM" }, { $set: { "location":"ERN" } });

```
MARKETING> db.EMPLOYEE.updateMany({ "location": "EKM" }, { $set: { "location": "ERN" } });
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
MARKETING> db.EMPLOYEE.find();
[
  {
    _id: ObjectId('666f0907b839dad0a1cdcdf6'),
    emp_id: 101,
    emp_name: 'SMITH',
    address: 'HNO:123',
    location: 'CLT'
  },
  {
    _id: ObjectId('666f0907b839dad0a1cdcdf7'),
    emp_id: 102,
    emp_name: 'ALLEN',
    address: 'HNO:456',
    location: 'ERN'
  },
  {
    _id: ObjectId('666f0907b839dad0a1cdcdf8'),
    emp_id: 103,
    emp_name: 'JONES',
    address: 'HNO:789',
    location: 'ERN'
  },
  {
    _id: ObjectId('666f0907b839dad0a1cdcdf9'),
    emp_id: 104,
    emp_name: 'JAMES',
    address: 'HNO:321',
    location: 'TVM'
  }
]
```

b) Display employee details in ascending order of their names.

> db.EMPLOYEE.find().sort({emp_name:1});

```
MARKETING> db.EMPLOYEE.find().sort({emp_name:1});
[
  {
    _id: ObjectId('666f0907b839dad0a1cdcdf7'),
    emp_id: 102,
    emp_name: 'ALLEN',
    address: 'HNO:456',
    location: 'ERN'
  },
  {
    _id: ObjectId('666f0907b839dad0a1cdcdf9'),
    emp_id: 104,
    emp_name: 'JAMES',
    address: 'HNO:321',
    location: 'TVM'
  },
  {
    _id: ObjectId('666f0907b839dad0a1cdcdf8'),
    emp_id: 103,
    emp_name: 'JONES',
    address: 'HNO:789',
    location: 'ERN'
  },
  {
    _id: ObjectId('666f0907b839dad0a1cdcdf6'),
    emp_id: 101,
    emp_name: 'SMITH',
    address: 'HNO:123',
    location: 'CLT'
  }
]
```

c) Display product names whose price greater than 25000.

> db.PRODUCT.find({ "price": { $gt : 25000 } });

```
MARKETING> db.PRODUCT.find({ "price": { $gt : 25000 } });
[
  {
    _id: ObjectId('666f09a7b839dad0a1cdcdfa'),
    p_id: 201,
    p_name: 'laptop',
    stock: 50,
    price: 40000
  },
  {
    _id: ObjectId('666f09a7b839dad0a1cdcdfd'),
    p_id: 204,
    p_name: 'smart watch',
    stock: 100,
    price: 30000
  }
]
```

EXPERIMENT NO: 11

AIM:
Create the following collections and documents in the database EMPMAPPING

Collection: EMPLOYEE

ssn	Ename	Address	Ph_no	department
301,	ALLEN	HNO:123	1234567891	DEP1
302,	SMITH	HNO:456	2345678912	DEP1
303,	JAMES	HNO:789	3456789123	DEP2
304,	JOHN	HNO:321	4567891234	DEP3

Collection: PROJECT

Pid	p_name	type	client_name
401,	BILLING SYSTEM,	WEB	BRICK
402,	AIRLINE,	ANDROID,	ALEN
403,	E_BOOK,	CLOUD,	ALEN
404,	SMART PARKING,	ANDROID	PETER

RESULT:

- Creating database
  EMPMAPPING > use
  EMPMAPPING;

```
test> use EMPMAPPING
switched to db EMPMAPPING
```

- Creating Collection EMPLOYEE and PROJECT and inserting data to the collections
  > db.createCollection("EMPLOYEE");
  > db.EMPLOYEE.insertMany([{"ssn":301,"E_name":"ALLEN","Address":"HNO:123", "Ph_no":1234567891,"department":"DEP1"},{"ssn":302,"E_name":"SMITH","Address":"HNO:456","Ph_no":23456789123,"department":"DEP1"},{"ssn":303,"E_name": "JAMES","Address":"HNO:789","Ph_no":3456789123,"deepartent":"DEP2"},{"ssn": 304,"E_name":"JOHN","Address":"HNO:321","Ph_no":4567891234,"department":" DEP3"}]);

```
EMPMAPPING> db.createCollection("EMPLOYEE");
{ ok: 1 }
EMPMAPPING> db.EMPLOYEE.insertMany([{"ssn":301,"E_name":"ALLEN","Address":"HNO:123","Ph_no":1234567891,"department":"DEP1"},{"ssn"
:302,"E_name":"SMITH","Address":"HNO:456","Ph_no":23456789123,"department":"DEP1"},{"ssn":303,"E_name":"JAMES","Address":"HNO:789"
,"Ph_no":3456789123,"department":"DEP2"},{"ssn":304,"E_name":"JOHN","Address":"HNO:321","Ph_no":4567891234,"department":"DEP3"}]);

{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('666f162119d89cc087cdcdf6'),
    '1': ObjectId('666f162119d89cc087cdcdf7'),
    '2': ObjectId('666f162119d89cc087cdcdf8'),
    '3': ObjectId('666f162119d89cc087cdcdf9')
  }
}
```

> db.createCollection("PROJECT");

> db.PROJECT.insertMany([{"p_id":401,"p_name":"BILLING SYSTEM","type":"WEB","client_name":"BRICK"},{"p_id":402,"p_name":"AIRLINE","type":"ANDROID","client_name":"ALLEN"},{"p_id":403,"p_name":"E_BOOK","type":"CLOUD","client_name":"ALEN"},{"p_id" :404,"p_name":"SMARTPARKING","type":"ANDROID","client_name":"PETER"}]);

```
EMPMAPPING> db.PROJECT.insertMany([{"p_id":401,"p_name":"BILLING SYSTEM","type":"WEB","client_name":"BRICK"},{"p_id":402,"p_name":
"AIRLINE","type":"ANDROID","client_name":"ALLEN"},{"p_id":403,"p_name":"E_BOOK","type":"CLOUD","client_name":"ALEN"},{"p_id":404,"
p_name":"SMART PARKING","type":"ANDROID","client_name":"PETER"}]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('666f173b19d89cc087cdcdfa'),
    '1': ObjectId('666f173b19d89cc087cdcdfb'),
    '2': ObjectId('666f173b19d89cc087cdcdfc'),
    '3': ObjectId('666f173b19d89cc087cdcdfd')
  }
}
```

EXPERIMENT NO: 12

AIM:

Write down the MongoDB queries for the following questions.

a) Update the house number of employee ‘MARTIN’

b) Display project details except ‘ANDROID’.

c) Sort the collection employee in descending order of their names.

RESULT :

a) Update the house number of employee ‘JOHN’

> db.EMPLOYEE.updateOne({"E_name":"JOHN"},{$set:{"Address":"HNO:301"}});

```
EMPMAPPING> db.EMPLOYEE.updateOne({ "E_name" :"JOHN" }, { $set: { "Address" :"HNO:301" } });
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
EMPMAPPING> db.EMPLOYEE.find();
[
  {
    _id: ObjectId('666f162119d89cc087cdcdf6'),
    ssn: 301,
    E_name: 'ALLEN',
    Address: 'HNO:123',
    Ph_no: 1234567891,
    department: 'DEP1'
  },
  {
    _id: ObjectId('666f162119d89cc087cdcdf7'),
    ssn: 302,
    E_name: 'SMITH',
    Address: 'HNO:456',
    Ph_no: 23456789123,
    department: 'DEP1'
  },
  {
    _id: ObjectId('666f162119d89cc087cdcdf8'),
    ssn: 303,
    E_name: 'JAMES',
    Address: 'HNO:789',
    Ph_no: 3456789123,
    department: 'DEP2'
  },
  {
    _id: ObjectId('666f162119d89cc087cdcdf9'),
    ssn: 304,
    E_name: 'JOHN',
    Address: 'HNO:301',
    Ph_no: 4567891234,
    department: 'DEP3'
  }
]
```

b) Display project details except ‘ANDROID’.

> db.PROJECT.find({ type: { $ne :"ANDROID" } });

```
EMPMAPPING> db.PROJECT.find({ type: { $ne :"ANDROID" } });
[
  {
    _id: ObjectId('666f173b19d89cc087cdcdfa'),
    p_id: 401,
    p_name: 'BILLING SYSTEM',
    type: 'WEB',
    client_name: 'BRICK'
  },
  {
    _id: ObjectId('666f173b19d89cc087cdcdfc'),
    p_id: 403,
    p_name: 'E_BOOK',
    type: 'CLOUD',
    client_name: 'ALEN'
  }
]
```

c) Sort the collection employee in descending order of their names.

> db.EMPLOYEE.find().sort({ "E_name": -1 });

```
EMPMAPPING> db.EMPLOYEE.find().sort({ "E_name": -1 });
[
  {
    _id: ObjectId('666f162119d89cc087cdcdf7'),
    ssn: 302,
    E_name: 'SMITH',
    Address: 'HNO:456',
    Ph_no: 23456789123,
    department: 'DEP1'
  },
  {
    _id: ObjectId('666f162119d89cc087cdcdf9'),
    ssn: 304,
    E_name: 'JOHN',
    Address: 'HNO:301',
    Ph_no: 4567891234,
    department: 'DEP3'
  },
  {
    _id: ObjectId('666f162119d89cc087cdcdf8'),
    ssn: 303,
    E_name: 'JAMES',
    Address: 'HNO:789',
    Ph_no: 3456789123,
    department: 'DEP2'
  },
  {
    _id: ObjectId('666f162119d89cc087cdcdf6'),
    ssn: 301,
    E_name: 'ALLEN',
    Address: 'HNO:123',
    Ph_no: 1234567891,
    department: 'DEP1'
  }
]
```

EXPERIMENT NO: 13

AIM:

Create the following collections and documents in the database MVRATING

Collection: MOVIE

mv_id	m_name	language	year
501,	AVATHAR	ENGLISH,	2009
502,	THE PRINCESS	ENGLISH,	2022
503,	HOME	MALAYALAM,	2021
504,	DON	THAMIL,	2022
505,	DIRTY DEEDS	ENGLISH	2002
506	QUARANTINE	ENGLISH	2008

Collection: ACTOR

act_id	act_name	gender
601,	SAM,	M
602,	JOEY KING	F
603,	INDRANS	M
604,	SURYAH	M

RESULT

- Creating database
  MVRATING > use
  MVRATING;

```
test> use MVRATING;
switched to db MVRATING
```

- Creating Collection MOVIE and ACTOR and inserting data to the collections
-

> db.createCollection("MOVIE");

> db.MOVIE.insertMany([{"mv_id":501, "mv_name":"AVATHAR", "language":"ENGLISH", " year":2009},{"mv_id":502, "mv_name":"THE PRINCESS", "language":"ENGLISH", "year":2022}, {"mv_id":503, "mv_name":"HOME", "language":"MALAYALAM", "year":2021}, {"mv_id":504, "mv_name":"DON", "language":"THAMIL", "year":2022}, {"mv_id":505, "mv_name":"DIRTY DEEDS", "language":"ENGLISH", "year":2002}, {"mv_id":506, "mv_name":"QUARANTINE", "language":"ENGLISH", "year":2008}]);

>

```
MVRATING> db.createCollection("MOVIE");
{ ok: 1 }
MVRATING> db.MOVIE.insertMany([{"mv_id":501,"mv_name":"AVATHAR","language":"ENGLISH","year":2009},{"mv_id":502,"mv_name":"THE PRIN
CESS","language":"ENGLISH","year":2022},{"mv_id":503,"mv_name":"HOME","language":"MALAYALAM","year":2021},{"mv_id":504,"mv_name":"
DON","language":"THAMIL","year":2022},{"mv_id":505,"mv_name":"DIRTY DEEDS","language":"ENGLISH","year":2002},{"mv_id":506,"mv_name
":"QUARANTINE","language":"ENGLISH","year":2008}]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('666f29131aa0905371cdcdf6'),
    '1': ObjectId('666f29131aa0905371cdcdf7'),
    '2': ObjectId('666f29131aa0905371cdcdf8'),
    '3': ObjectId('666f29131aa0905371cdcdf9'),
    '4': ObjectId('666f29131aa0905371cdcdfa'),
    '5': ObjectId('666f29131aa0905371cdcdfb')
  }
}
```

> db.createCollection("ACTOR");

> db.ACTOR.insertMany([{"act_id":601, "act_name":"SAM", "gender":"M"}, {"act_id":602,"act_name":"JOEY KING", "gender":"F"}, {"act_id":603, "act_name":"INDRANS", " gender":"M"}, {"act_id":604, "act_name":"SURYAH", "gender":"M"}]);

```
MVRATING> db.createCollection("ACTOR");
{ ok: 1 }
MVRATING> db.ACTOR.insertMany([{"act_id":601,"act_name":"SAM","gender":"M"},{"act_id":602,"act_name":"JOEY KING","gender":"F"},{"a
ct_id":603,"act_name":"INDRANS","gender":"M"},{"act_id":604,"act_name":"SURYAH","gender":"M"}]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('666f29d11aa0905371cdce00'),
    '1': ObjectId('666f29d11aa0905371cdce01'),
    '2': ObjectId('666f29d11aa0905371cdce02'),
    '3': ObjectId('666f29d11aa0905371cdce03')
  }
}
```

EXPERIMENT NO: 14

AIM:

Write down the MongoDB queries for the following questions.

a) Update the releasing year of DON as 2021.

b) Display details of movies in the year of release 2022.

c) Display sorted movie details in order of their releasing date.

RESULT :

a) Update the releasing year of DON as 2021.

> db.MOVIE.updateOne({ "mv_name": "DON" }, { $set: { "year": 2021 } });

```
MVRATING> db.MOVIE.updateOne({ "mv_name": "DON" }, { $set: { "year": 2021 } });
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
MVRATING> db.MOVIE.find();
[
  {
    _id: ObjectId('666f29131aa0905371cdcdf6'),
    mv_id: 501,
    mv_name: 'AVATHAR',
    language: 'ENGLISH',
    year: 2009
  },
  {
    _id: ObjectId('666f29131aa0905371cdcdf7'),
    mv_id: 502,
    mv_name: 'THE PRINCESS',
    language: 'ENGLISH',
    year: 2022
  },
  {
    _id: ObjectId('666f29131aa0905371cdcdf8'),
    mv_id: 503,
    mv_name: 'HOME',
    language: 'MALAYALAM',
    year: 2021
  },
  {
    _id: ObjectId('666f29131aa0905371cdcdf9'),
    mv_id: 504,
    mv_name: 'DON',
    language: 'THAMIL',
    year: 2021
  },
  {
    _id: ObjectId('666f29131aa0905371cdcdfa'),
    mv_id: 505,
    mv_name: 'DIRTY DEEDS',
    language: 'ENGLISH',
    year: 2002
  },
  {
    _id: ObjectId('666f29131aa0905371cdcdfb'),
    mv_id: 506,
    mv_name: 'QUARANTINE',
    language: 'ENGLISH',
    year: 2008
  }
]
```

b) Display details of movies in the year of release 2022.

> db.MOVIE.find({ "year": 2022 })

```
MVRATING> db.MOVIE.find({ "year": 2022 })
[
  {
    _id: ObjectId('666f29131aa0905371cdcdf7'),
    mv_id: 502,
    mv_name: 'THE PRINCESS',
    language: 'ENGLISH',
    year: 2022
  }
]
```

c) Display sorted movie details in order of their releasing date.

> db.MOVIE.find().sort({ "year": 1 })

```
MVRATING> db.MOVIE.find().sort({ "year": 1 })
[
  {
    _id: ObjectId('666f29131aa0905371cdcdfa'),
    mv_id: 505,
    mv_name: 'DIRTY DEEDS',
    language: 'ENGLISH',
    year: 2002
  },
  {
    _id: ObjectId('666f29131aa0905371cdcdfb'),
    mv_id: 506,
    mv_name: 'QUARANTINE',
    language: 'ENGLISH',
    year: 2008
  },
  {
    _id: ObjectId('666f29131aa0905371cdcdf6'),
    mv_id: 501,
    mv_name: 'AVATHAR',
    language: 'ENGLISH',
    year: 2009
  },
  {
    _id: ObjectId('666f29131aa0905371cdcdf8'),
    mv_id: 503,
    mv_name: 'HOME',
    language: 'MALAYALAM',
    year: 2021
  },
  {
    _id: ObjectId('666f29131aa0905371cdcdf9'),
    mv_id: 504,
    mv_name: 'DON',
    language: 'THAMIL',
    year: 2021
  },
  {
    _id: ObjectId('666f29131aa0905371cdcdf7'),
    mv_id: 502,
    mv_name: 'THE PRINCESS',
    language: 'ENGLISH',
    year: 2022
  }
]
```